



A discrete Keplerian optimization algorithm with Q-learning for human-robot collaborative partial disassembly line balancing problem

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HIGHLIGHTS

- Human-Robot collaborative partial disassembly is considered in DLBP.
- An improved Keplerian optimization algorithm is proposed for HRC-PDLBP.
- A Q-learning strategy is presented to adjust the local search operators.
- The effectiveness of the algorithm is verified by comparison experiments.

ARTICLE INFO

Keywords:

Partial disassembly line balancing
Human-robot collaborative
Discrete Kepler optimization algorithm
Q-Learning

ABSTRACT

Driven by advancements in human-machine collaboration technology, waste recycling enterprises are increasingly pursuing innovative production methodologies. Nevertheless, in practical human-machine collaborative partial disassembly tasks, task variability, process uncertainty, and multi-objective conflicts persist, rendering existing approaches inadequate for balancing efficiency, economic performance, and energy consumption. To address these challenges, this paper introduces a discrete Kepler optimization algorithm integrated with Q-learning (QLS-KOA) to resolve the human-machine collaborative disassembly line balancing problem (HRC-PDLBP), with the objective of simultaneously optimizing load distribution, disassembly profit, and energy consumption. At first, a two-dimensional coding scheme is designed to transform the continuous Kepler optimization algorithm into a discrete form, thereby generating high-quality initial solutions well-suited to the problem's characteristics. Then, a Q-learning based local search framework is proposed to dynamically adjust the proportion of four HRC-PDLBP-specific local search operators in response to variations in the solution state. Considering that the solution space of HRC-PDLBP can dynamically change with task attributes and constraints, a Q-learning local search framework is introduced to make the intelligent agents efficiently capture optimization directions, discover global optimal solutions, and avoid local optima in dynamic environments. Finally, comparative experiments on multiple benchmark instances indicate that QLS-KOA achieves higher solution accuracy, faster convergence, and better solution diversity than state-of-the-art algorithms, while simultaneously reducing energy consumption and improving disassembly profit. Furthermore, a real case study on Tesla Model S battery module confirms that QLS-KOA generates practical and efficient disassembly schemes, and demonstrates significant advantages in terms of overall performance, sustainability, and industrial applicability.

1. Introduction

In recent decades, the continuous development of global technology has accelerated the obsolescence of electronic products, shortened their lifespans and resulted in an explosive increase in electronic waste. This has led to significant resource waste and environmental hazards.

Effectively utilizing and recycling End-of-Life (EOL) products has become a consensus among organizations worldwide. The disassembly line, as one of the most critical stages in the remanufacturing, reuse, and recycling of discarded products, has been attracted considerable attention from recycling enterprises. Market competition has prompted companies to increasingly prioritize the efficiency and economic

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<https://doi.org/10.1016/j.asoc.2025.114406>

Received 3 April 2025; Received in revised form 6 November 2025; Accepted 30 November 2025

Available online 9 December 2025

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advantages of disassembly lines. Partial disassembly, in comparison to full disassembly, has gained traction among companies due to its selective recovery of high-value components, reduction of resource wastage, revenue increment, and enhancement of disassembly efficiency. In recent years, the issue of partial disassembly line balancing has emerged as a prominent focus in the scheduling domain [4].

Traditional disassembly lines predominantly rely on manual labor, yet they incur high costs and exhibit inefficiency. Furthermore, the presence of hazardous components in discarded products poses health risks. With advancements in robotics and automation, robotic disassembly has gradually supplanted manual labor, enhancing efficiency. However, robots face limitations in executing complex tasks. Consequently, human-robot collaborative disassembly systems amalgamate the adaptability of human operators with the efficacy of robots to ensure optimal efficiency and operator safety, thereby augmenting productivity and bolstering company profitability. Moreover, within a collaborative disassembly line involving humans and machines, considerations should extend beyond mere productivity and profit. The energy consumption of equipment and the workload of personnel demand simultaneous attention. Specifically, hazardous and attribute-less parts are assigned to machines, while complex attributes are delegated to humans. This reduces the burden on personnel and ensures their safety and efficiency. Additionally, an increasing number of dismantling companies are opting for partial dismantling, which involves disassembling only valuable parts to reduce unnecessary energy consumption and increase revenue. However, this also heightens the complexity of the problem due to the uncertainty surrounding the depth of disassembly. To remedy this deficiency, this paper presents a human-machine collaborative partial disassembly line balancing problem (HRC-PDLBP), which integrates the multiple influencing factors of the real disassembly problem from the perspective of designing a complete production line targeting profit, energy consumption, and production smoothing, and combines the various costs of actual production, including the cost of opening the station, the cost of purchasing the robot, and the operation cost.

In addition, since HRC-PDLBP entails addressing partially disassembled DLBP within human-machine collaboration contexts, it encompasses the uncertainty surrounding detachable components and the constraints regarding the disassembly of allocated parts, there exists a potential for the solving algorithm to converge to a locally optimal solution. To address this potential local optimality issue and enhance the algorithm's adaptability to the dynamically changing solution states of HRC-PDLBP, this study amalgamates Q-learning with the Discrete Keplerian Optimization Algorithm (KOA) to devise a novel approach termed Q-learning Local Search-Discrete Keplerian Optimization Algorithm (QLS-KOA). In QLS-KOA, the KOA algorithm is discretized to align with the characteristics of HRC-PDLBP. Additionally, a local search framework based on Q-learning is introduced. This framework distinguishes itself by its capacity to dynamically adjust the proportion of four local search operators in response to variations in the solution state. It encompasses the selection of local search actions by the intelligent agent, feedback from the environment's reward mechanism, and the setting of the corresponding problem-solving state. Such adaptable adjustments empower the intelligent agent to efficiently discover the optimal solution amidst evolving problem environments, with the objective of addressing the HRC-PDLBP problem and enhancing the algorithm's search efficiency and accuracy. The main contributions of this paper are as follows.

- (1) This paper presents a human-robot collaborative partial disassembly line balancing problem and formulates the corresponding mathematical model.
- (2) A novel discrete KOA algorithm with Q-learning is proposed to solve the problem, in which a dynamic proportional local search framework based on Q-learning is proposed to adjust the ratio of the four local search operators based on the solution state, aiming

to improve the efficiency of the algorithm in exploring the solution space.

- (3) The effectiveness and efficiency of the proposed model and method were validated through numerical experiments, and further confirmed through a real-world disassembly case of a Tesla Model S battery module, demonstrating its practical applicability.

The remaining sections of this study are outlined as follows: [Section 2](#) explore the research background, optimization methods, and research motivations related to the production line balancing problem. The introduced HRC-PDLBP is outlined in [Section 3](#). [Section 4](#) elucidates the methodological details of QLS-KOA for tackling HRC-PDLBP. [Section 5](#) is designed to conduct a series of comparative experiments to assess the exceptional performance exhibited by the proposed QLS-KOA. Moreover, [Section 6](#) encapsulates the primary research findings and delineates prospective avenues for further investigation.

2. Related works

The disassembly line balancing problem is a critical stage in the remanufacturing process [11,40]. So many experts and researchers have delved into this problem, and initially, a large body of studies focused on manual disassembly [28,31,38]. However, with the advancement of automation technology, increasing research endeavors now prioritize efficient robotic disassembly to enhance productivity [10,39,42]. Nevertheless, robotic disassembly encounters constraints, especially when confronted with intricate disassembled components, leading to potential inefficiencies in task completion by machinery. Consequently, a growing body of scholars is delving into the exploration of human-machine collaborative disassembly models. Nonetheless, this remains a challenge due to the complexity of allocation rules and coordination between humans and machines.

In addressing productivity, Xu et al. [36] endeavored to minimize the demand index while optimizing workstation count and job smoothing in a complete disassembly line scenario with a predefined cycle time. They also factored in the safety of operator-assisted machinery during task execution. Similarly, Jin et al. [15] concentrated on cycle time, robot count, and labor force allocation in another complete disassembly line setup with a fixed workstation count. On the economic front, Wu et al. [35] considered not only workstation count and job smoothing, but also the disassembly process costs within a specified cycle time. Additionally, Zhao et al. [43] expanded the scope by incorporating disassembly revenue, defining profit as revenue minus total costs, thereby offering a more precise depiction and optimization of the problem. With the increasing emphasis on sustainable development, scholars have also begun to pay attention to energy consumption in the production process. Fang et al. [6] introduced a complete disassembly model based on positional constraints with a fixed number of workstations, which considers cycle time, manpower, and robot energy consumption as the objectives. Guo et al. [9] proposed a human-robot collaborative given cycle time under stochastic task time completely disassembly line equilibrium model that integrates profit and energy consumption optimization. The above studies primarily concentrate on complete disassembly. However, in practical application, disassembling only valuable components becomes imperative to mitigate unnecessary losses and energy consumption associated with dismantling non-valuable parts. The partial disassembly paradigm offers a means to effectively curtail resource wastage and enhance profitability. Nevertheless, the challenge is compounded by the uncertainty surrounding component dismantlement in partial disassembly scenarios. To date, no investigation has integrated these dual aspects. Therefore, this study presents for the first time a human-machine collaborative partial disassembly production line and simultaneously optimizes profit, energy consumption, and production smoothing from different optimization perspectives such as economic, environmental, and productivity.

Table 1
Summary of the literature review.

Related works	Operator			Problem type		Mode		Solving methods
	Manual	Robotic	Collaboration	complete	partial	Type I	Type II	
Yin et al. [38]	●			●			●	INSGA-II
Tuo et al. [28]	●					●		SSA-DE
Wang et al. [31]	●			●		●		GSA
Guo et al. [10]		●		●			●	ENHABC
Li et al. [18]		●		●			●	VNS+RL
Yang et al. [37]	●			●		●		MOTLBO +RL
Ren et al. [21]		●		●		●		VNIS+RL
Zeng et al. [42]		●			●		●	IGSA
Xu et al. [36]			●	●		●		IDBA
Jin et al. [15]			●	●			●	B-DMOEA
Wu et al. [35]			●	●		●		HLSGA
Guo et al. [8]			●	●		●		VND-ABC
Fang et al. [6]			●	●			●	MO-MFO
Wang et al. [33]		●		●		●		NSGA-II+RL
Ren et al. [22]		●		●			●	ABC+RL
This study			●		●	●		KOA+RL

* Type I: The cycle time is given. Type II: The number of workstations is given.

* RL: Reinforcement Learning.

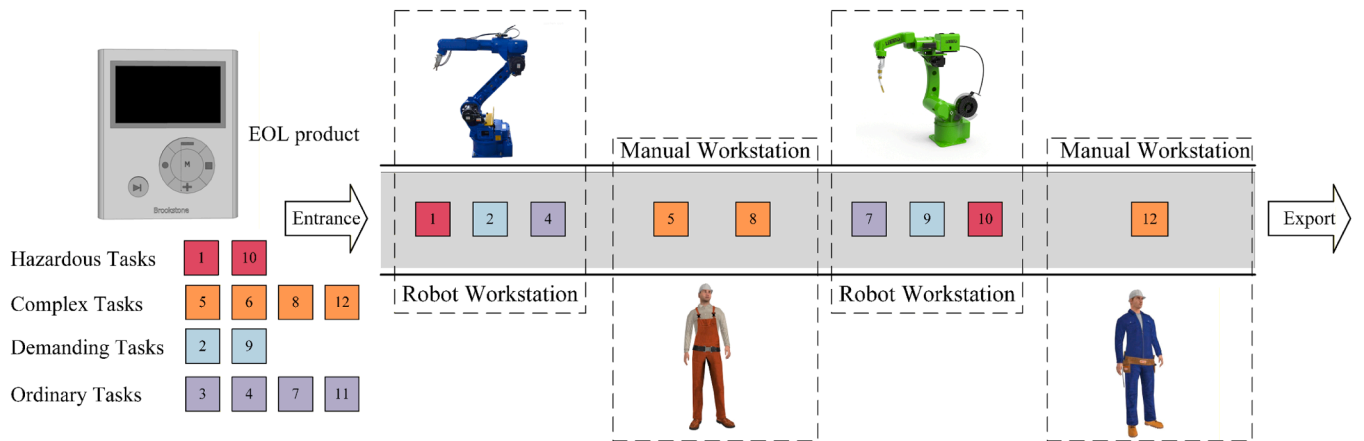


Fig. 1. Diagram of the human-machine collaborative partial disassembly line.

The DLBP has been established as a classical NP-complete problem [32], while the HRC-PDLBP presents a heightened level of intricacy. Genetic algorithm [17], ant colony optimization algorithm [5], northern goshawk optimization algorithm [27], discrete bee colony algorithm [8], genetic simulated annealing algorithm [41], scatter search algorithm [12], discrete flower pollination algorithm [30], and other meta-heuristic algorithm are extensively employed to address DLBP issues due to their robust search capabilities, ease of manipulation, and high precision. Among them, KOA, as a newly proposed meta-heuristic algorithm [1] with high flexibility, easy tuning and better parallelization, has successfully solved many combinatorial optimization problems [13,2]. Notably, it boasts a simple yet robust structure, exhibits strong global search capabilities, and ensures greater stability compared to alternative metaheuristic algorithms. Compared with traditional meta-heuristic algorithms that often rely on fixed crossover/mutation mechanisms or pheromone update rules-resulting in limited exploration of the complex solution space for HRC-PDLBP. KOA mimics Kepler's laws of planetary motion to drive continuous expansion and adaptive adjustment of the search space, enabling more comprehensive coverage of HRC-PDLBP's discrete, multi-constraint solution domain and better adaptability to its problem complexity. This study represents the first application of the KOA algorithm to the optimization of DLBP, thereby introducing a novel application context for this method in discrete optimization and establishing a new heuristic paradigm for addressing complex human-machine collaborative disassembly line balancing

problems.

Additionally, when addressing more intricate disassembly line balancing challenges, KOA encounters the difficulty of adjusting to frequently changing solution states. Integrating the Q-learning algorithm with KOA can harness the adaptability of the Q-learning algorithm to changing solution states and utilize the optimization capabilities of metaheuristics to formulate efficient disassembly plans. Existing literatures have already explored the methods of combining intelligent algorithms and Q-learning to solve the DLBP-related problems. However, these studies mainly focus on employed Q-learning solely as an operator selection mechanism, selecting a single local search operator from a predefined pool during iterations [18,21,22,33,37]. Since the absence of dynamic control over multi-operator combinations and their relative proportions restricts search diversity and global exploration, they often cause the algorithm to converge prematurely to local optima. To overcome this limitation, this paper proposes a Q-learning driven mechanism to dynamically adjust the proportions of multiple neighborhood search operators in response to feedback from the optimization process. It enhances search diversity, mitigates premature convergence, and enables flexible adaptation to the varying characteristics of disassembly problems across different stages. This integrated strategy is anticipated to effectively tackle the complexity of HRC-PDLBP, which takes advantage of the flexibility and generalization ability of reinforcement learning to automate the decision-making process in complex environments and tasks, which dynamically adjust the search ratio of the four search

operators according to the solution state of the problem, so as to ensure its effectiveness in solving HRC-PDLBP. A review of the above studies is presented in Table 1.

3. Problem description and mathematical modeling

3.1. Problem description

Compared with the traditional DLBP method, HRC-PDLB is more complex and requires the reasonable allocation of tasks based on the constraints of the task attributes. In this paper, end-of-life (EOL) parts are categorized into four different categories: hazardous, complex, high-demand, and conventional. As illustrated in Fig. 1, the disassembly line allocates hazardous parts to robotic workstations, while complex parts unsuitable for automated disassembly are manually handled. Conventional parts are assigned to robotic workstations under the constraints of a formulized model. This assignment alleviates the manual workload, enhances worker adaptability, and reduces the overall cost of disassembly tasks. The disassembly line tasks comprise the extraction of hazardous and high-demand parts, as well as the random selection of remaining parts for disassembly under relevant constraints. The objectives are to achieve optimal workload balance, improve profitability, and minimize energy consumption. With respect to disassembly costs, in line with practical production scenarios, the activation costs of workstations, robot procurement expenses, and respective operational costs of both robotic and manual workstations are taken into account.

The HRC-PDLB contains two sub-issues. Firstly, there is the issue of rationally selecting the parts for disassembly, noting that not all components in partial disassembly require disassembly. Secondly, there is the task of judiciously assigning disassembly tasks to either robots or humans to ensure equitable distribution of workload across workstations and in accordance with constraints. Consequently, this study integrates the selection of disassembly components and task allocation to yield a reasonable disassembly solution. To facilitate the construction of the mathematical model, this paper provides the following assumptions.

- (1) There exists only a singular product type on the disassembly line.
- (2) The precedence relations between disassembly tasks for each task are known.
- (3) The available data encompasses the operational duration and earnings associated with each task, along with the unitary cost in terms of time and energy consumption for both the robotic and human elements.
- (4) Disassembly tasks with hazards, complex and demands are known. And the hazards and demands tasks must be disassembled.

3.2. Mathematical programming

The established mathematical model is as follows, and the parameters and variables required by the model are as follows:

Indices:	
i, j	Index of disassembly task; $i, j \in I = \{1, 2, \dots, n\}$
k, l	Index of workstation; $k, l \in M = \{1, 2, \dots, m\}$
n	The quantity of disassembly tasks
m	The quantity of available workstations
I	The collection of all disassembly tasks, $I = \{1, 2, \dots, n\}$
M	The collection of all workstations, $M = \{1, 2, \dots, m\}$
B_c	The collection of tasks with complex attributes, known quantities
B_h	The collection of tasks with hazardous attributes, known quantities
$P_{AND}(i)$	The task set that immediately comes before task i and fulfills the AND priority relationship
$P_{OR}(i)$	The task set that immediately comes before task i and fulfills the OR priority relationship
$P(i)$	Set of all predecessors of disassembly task i
t_i	Disassembly time of task i
ρ_i	Disassembly profit of disassembly task i
ST_i	Start time of task i
C_w	Expenditure to initiate a workstation

(continued on next column)

(continued)

Indices:	
C_r	Expenditure to buy a robot
C_{pt}	The expenditure per unit time of worker disassembly
C_{rt}	The expenditure per unit time of robot disassembly
E_{pt}	The energy consumption of worker disassembly
E_{rt}	The energy consumption of robot disassembly
CT	Cycle time of disassembly line
AT	Actual production cycle time
Ψ	A large positive number
Decision variables:	
x_{ik}	If task i is assigned to the workstation k , then $y_{ik} = 1$; otherwise $y_{ik} = 0$
y_{ijk}	If both task i and task j are designated to the identical workstation k , and if task i is disassembled before task j , then $y_{ijk} = 1$; otherwise, $y_{ijk} = 0$
S_k	Workstation open variable, $S_k = 1$ if workstation k is open; otherwise $S_k = 0$
R_k	Robot workstation open variable, where $R_k = 1$ if workstation k is a robot workstation; otherwise, $R_k = 0$
P_k	Manual workstation open variable, where $P_k = 1$ if workstation k is a manual workstation; otherwise, $P_k = 0$
Auxiliary variables:	
h_i	$h_i = 1$ if task i is hazardous; otherwise $h_i = 0$
d_i	$d_i = 1$ if task i is demanding; otherwise $d_i = 0$

3.2.1. Objective functions and constraints

The following describes the goals and constraints of the proposed HRC-PDLB:

$$f_1 = \sqrt{\sum_{k=1}^m \left(S_k \bullet AT - \sum_{i=1}^n x_{ik} \bullet t_i \right)^2} \# \quad (1)$$

$$f_2 = \sum_{k=1}^m \sum_{i=1}^n x_{ik} \rho_i - \sum_{k=1}^m \left[(C_w \bullet S_k + C_r \bullet R_k) + \sum_{i \in I} (C_{pt} \bullet P_k + C_{rt} \bullet R_k) \bullet x_{ik} \bullet t_i \right] \# \quad (2)$$

$$f_3 = \sum_{k=1}^m \sum_{i \in I} (E_{pt} \bullet P_k + E_{rt} \bullet R_k) \bullet x_{ik} \bullet t_i \# \quad (3)$$

$$\sum_{k=1}^m (h_i \bullet x_{ik} + d_i \bullet x_{ik}) \geq 1 \quad \forall i \in \{i | h_i + d_i \geq 1\} \# \quad (4)$$

$$\sum_{k=1}^m (h_i \bullet x_{ik} + d_i \bullet x_{ik}) \leq 1 \quad \forall i \in \{i | h_i + d_i = 0\} \# \quad (5)$$

$$x_{ik} \bullet P_k = 1 \quad \forall i \in B_c \# \quad (6)$$

$$x_{ik} \bullet R_k = 1 \quad \forall i \in B_h \# \quad (7)$$

$$x_{il} \leq \sum_{k=1}^m x_{jk}, \quad \forall i \in I; \forall j \in P_{AND}(i) \# \quad (8)$$

$$x_{il} \leq \sum_{k=1}^m x_{jk}, \quad \forall i \in I; \exists j \in P_{OR}(i) \# \quad (9)$$

$$\sum_{i=1}^n x_{ik} \bullet t_i \leq CT \bullet S_k, \quad \forall k \in M \# \quad (10)$$

$$AT = \max \left(\sum_{i \in I} x_{ik} \bullet t_i \right) \# \quad (11)$$

$$CT \bullet \left(\sum_{k \in M} x_{ik} \bullet k - 1 \right) + \sum_{k \in M} \sum_{i, j \in I} y_{ijk} \bullet t_i \leq ST_j, \quad \forall i, j \in I; i \neq j \# \quad (12)$$

$$\Psi \bullet (1 - y_{ijk}) + ST_i \geq ST_j + t_i, \quad \forall i, j \in I; i \neq j; \forall k \in M \# \quad (13)$$

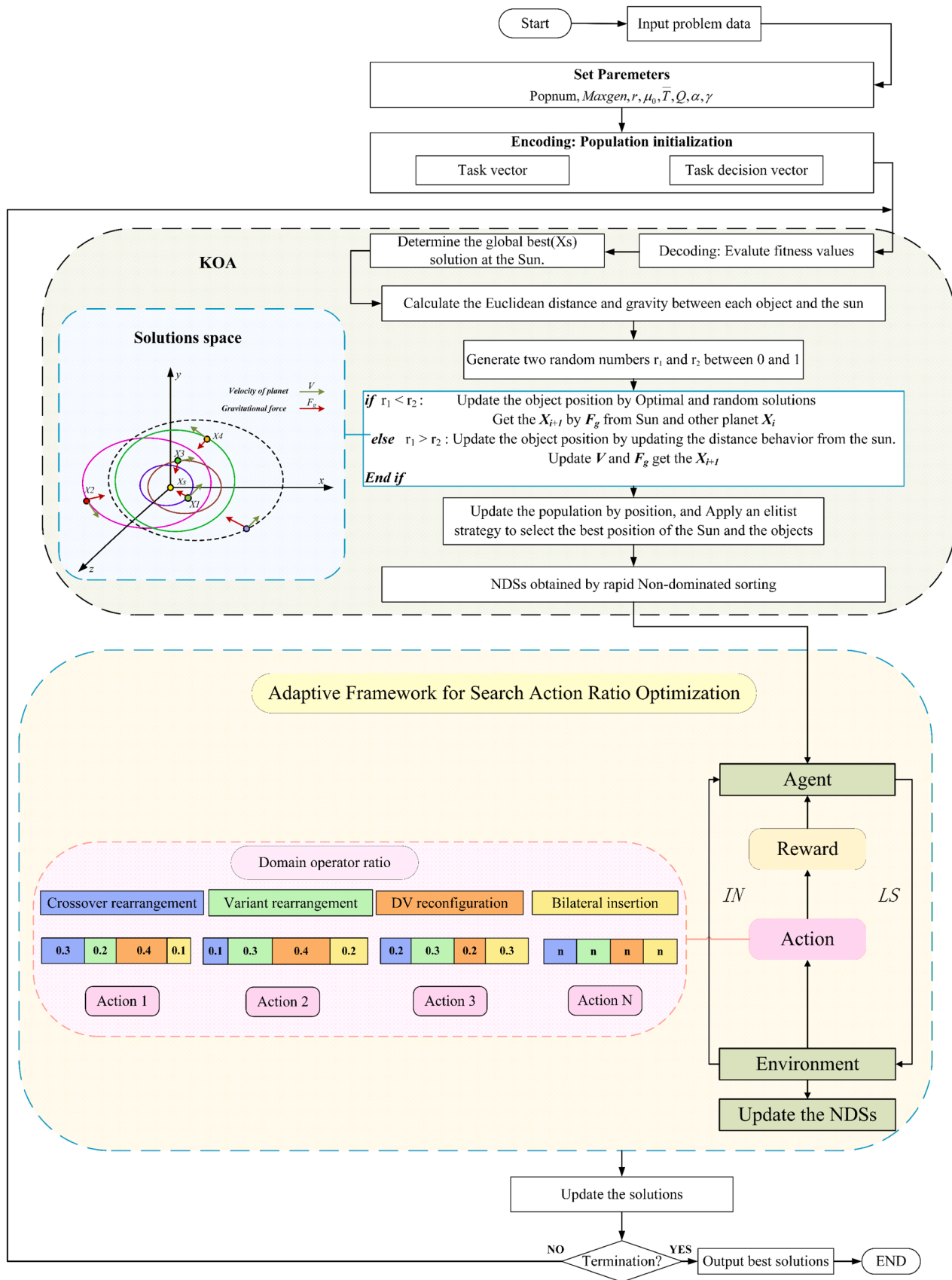


Fig. 2. The flowchart of the proposed QLS-KOA.

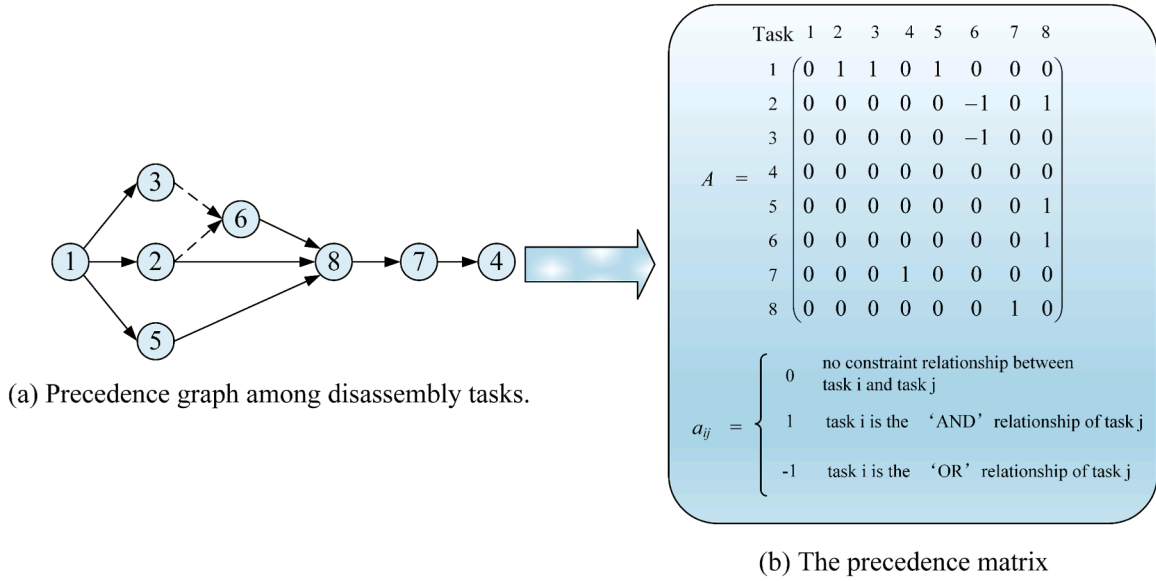


Fig. 3. An example of a simple PC with 8 components.

$$ST_i \geq 0, \forall i \in I\# \quad (14)$$

$$x_{ik} + x_{jk} \leq 1 + (y_{ijk} + y_{jik}), \forall i, j \in I; i < j; \forall k \in M\# \quad (15)$$

$$x_{ik} + x_{jk} \geq 2 \cdot (y_{ijk} + y_{jik}), \forall i, j \in I; i < j; \forall k \in M\# \quad (16)$$

Eq. (1) is designed to augment production efficiency through the optimization of load balancing across workstations. Eq. (2) encapsulates the maximization of disassembly profit, calculated as the difference between revenue and costs. The costs associated with the disassembly station encompass the establishment and operational expenses of the station and the acquisition of robots. Eq. (3) articulates the minimization of energy consumption in the disassembly line, incorporating the energy utilization of robot workstations and human workstations. The constraints stipulate that Eq. (4) mandates the disassembly of parts with hazards and requisites. Eq. (5) posits that components lacking both hazard and significance may be either selected for disassembly or left intact. Eq. (6) dictates that intricate tasks are executed by human agents, while Eq. (7) stipulates that hazardous tasks are undertaken by robots. Eq. (8) prescribes the conditions for tasks under the AND precedence, wherein a task cannot be selected until the completion of all its preceding tasks, and assignment to a workstation currently in operation is prohibited. Eq. (9) ensures that for tasks under the OR precedence, at least one of its predecessors must be finalized before it becomes available for selection. The duration of a task within a workstation, as limited by Eqs. (10) and (11), ensures that it does not exceed the allocated cycle time to avoid constraints. Eq. (12) specifies the parameters of the actual production cycle time. Eqs. (13) and (14) assert that tasks allocated to the same workstation must be carried out sequentially, with each succeeding task commencing only after the completion of its antecedent. Eqs. (15) and (16) embody location-based constraints, asserting that tasks i and j , when designated to the same workstation, are precluded from allocation to any alternative workstations.

4. The proposed QLS-KOA for HRC-PDLBP

As illustrated in Fig. 2, the QLS-KOA consists of two functional components: a targeted two-dimensional coding scheme is first constructed to initialize the population, where each celestial body represents a candidate solution by encoding task assignment sequences and workstation allocation information. It addresses the defect of traditional

static coding that fails to fully reflect the coupling relationship between tasks and resources. During the optimization process, the algorithm dynamically refines the positional information of celestial bodies to generate new populations (NPs), with each individual adjusting its relative position to the Sun (i.e., the current optimal solution) at adaptive intervals. Unlike conventional static update strategies, the positional information of celestial bodies is randomly updated during iteration, thereby enhancing the diversity of the solution space while ensuring directional convergence toward the global optimal solution. Additionally, an improved elite retention strategy is incorporated to efficiently preserve top-performing individuals and screen non-dominated solutions (NDSs), avoiding the loss of high-quality solutions caused by random perturbations in the optimization process. The QLS-KOA further integrates Q-learning into the celestial body optimization framework to realize dynamic adaptive search of the solution space. It enables the algorithm to learn the value of different search directions through Q-value iteration, dynamically adjusting the search step size and scope based on the feedback of solution quality and overcoming the insufficient local search capability of traditional celestial body optimization algorithms. The iterative process alternates between celestial body position updates and Q-learning-driven adaptive search, terminating when a preset convergence criterion is satisfied, and ultimately yielding a set of high-quality optimal NDSs with better balance between diversity and convergence.

4.1. Encoding

In the PDLBP, a precedence diagram is frequently employed to illustrate the interdependencies among components in disassembled products. For instance, Fig. 3 delineates the precedence relationships among eight disassembly tasks and the mathematical matrix.

Fig. 3(a) illustrates the priority relationships disassembly tasks. Based on it, Fig. 3(b) illustrates the specific construction process of the precedence matrix A . The element a_{ij} in the matrix (where i and j denote task indexes) signifies the AND/OR prioritization relationship among decomposed tasks. Drawing upon the attributes of partial disassembly lines, a two-dimensional coding approach, denoted as [TV, DV], has been devised for individual representation. Here, TV signifies the task sequence vector, and DV denotes the operation decision vector. The detailed encoding procedures are outlined as follows:

Step1: Tasks with zero entries in the priority matrix are designated as candidates at the present position.

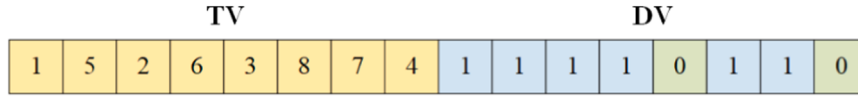


Fig. 4. Coding sequence diagram.

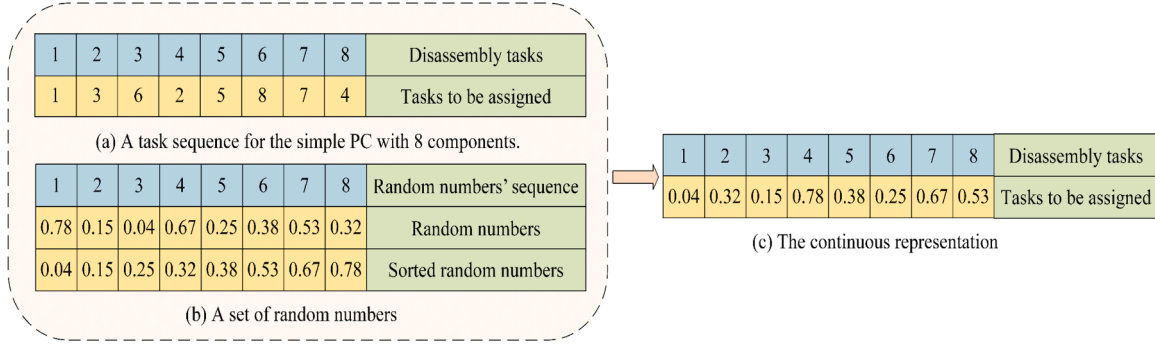


Fig. 5. Discrete operation.

Step2: Randomly select a task from the candidate task set as the task for the current position.

Step3: Remove the precedence links associated with the selected task and update the priority matrix, then update the priority matrix.

Step4: Repeat steps 1–3 until all tasks are allocated. Get the TV vector

Step5: The demanding and hazardous tasks, along with their predecessors, are designated as 1, whereas the disassembly variables for the remaining tasks are randomly determined from the set $\{0, 1\}$; Get the DV vector

Step6: Obtain the initialization encoding.

Fig. 4 presents a feasible disassembly sequence obtained from eight disassembly tasks involving discarded components. The sequencing between Task 2 and Task 6 (or between Task 7 and Task 8) adheres to the sequential requirements. Furthermore, the disassembly variable indicates that Task 3 and Task 4 do not necessitate disassembly.

Given that the disassembly line balancing problem entails a discrete combination optimization challenge, the applicability of the KOA method diminishes. Consequently, this study converts the discrete representation of disassembly task arrangements into a continuous format. Taking the disassembly tasks from Fig. 3 as an example of task arrangement, illustrated in Fig. 5. (a). To represent this continuous ordering, eight random numbers are initially generated within the range of $(0, 1)$, corresponding to the number of tasks to be ordered. Subsequently, these numbers are sorted to construct a position vector, as depicted in Fig. 5. (b). For the creation of a continuous representation, the positioning of the random numbers within the position vector must align with the sequence of the designated disassembly tasks. As outlined in Fig. 5. (c), the assignments adhere to ascending order, where the smallest random number corresponds to disassembly task 1, followed by subsequent assignments in ascending order to the respective tasks. Specifically, the position of the smallest value within the position vector corresponds to the first element of the task sequence vector, and this sequence continues accordingly.

4.2. Decoding

The decoding operators aim to transform a solution into a particular task scheme, adhering to both cycle time constraints and task attribute

constraints. The sequence of task execution is delineated by the vector TV, while the vector DV dictates the execution status of tasks. This paper employs a decoding method rooted in task attributes, allocating tasks with intricate and hazardous attributes to human-operated and robotic workstations, respectively. Tasks lacking specific attributes are assigned based on the minimization of model objectives. Within the realm of partial disassembly line balancing problems, the strategic allocation of tasks deemed hazardous or high in demand, along with their antecedents, is crucial. Multiple possibilities arise concerning the allocation of other tasks. Consequently, diverse solutions can be derived through distinct allocation principles. The particular decoding process is delineated in Fig. 6.

4.3. Initialization

Within the QLS-KOA framework, each planet symbolizes a potential solution, defined by unique positions and velocities. The corresponding parameters are adjusted according to the laws of planetary motion. The optimal solution is subsequently designated as the 'sun,' determined by the states of the candidate solutions. The proposed algorithm generates a random distribution of N planets (population size) across d dimensions, where d represents the dimensionality of the optimization problem under consideration. This is expressed by the following equation:

$$X_i^j = \gamma \bullet X_{i,up}^j + X_{i,low}^j \bullet (1 - \gamma), i = 1, 2, \dots, N; j = 1, 2, \dots, \text{dim\#} \quad (17)$$

Where N represents the number of populations, dim indicates the dimensionality of the solution space, X_i^j denotes the i th candidate solution, $X_{i,up}^j$ and $X_{i,low}^j$ represent the upper and lower bounds of the dimensions pertinent to the problem being dealt with, respectively; γ signifies a random number within the range of 0 and 1.

4.4. Calculating an object's velocity

The velocity of an object is defined by its proximity to the sun, increasing when nearer and decreasing when farther due to the sun's gravitational influence, which is stronger at close range and weaker at greater distances. This phenomenon is mathematically represented by

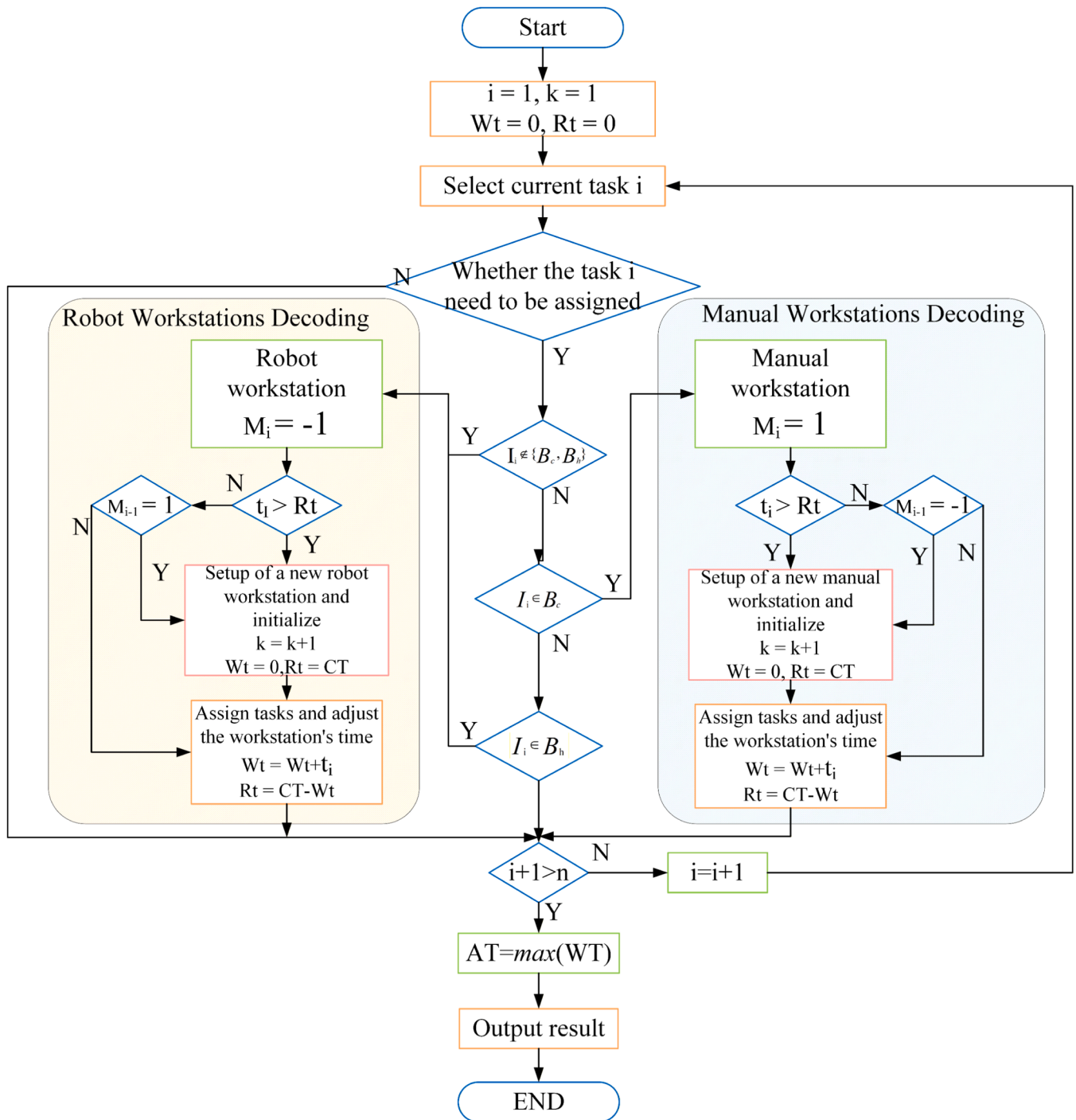


Fig. 6. Decoding flowchart.

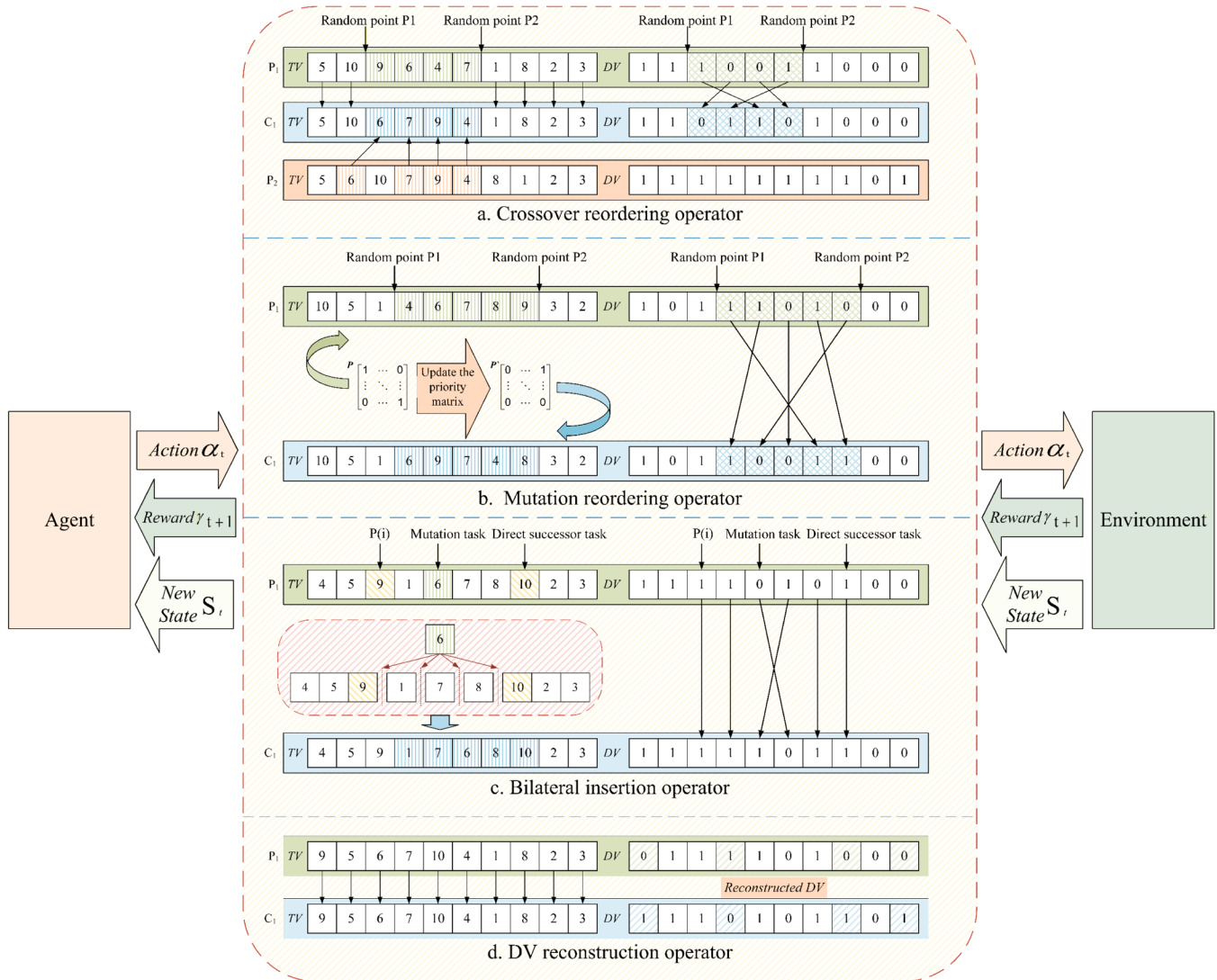
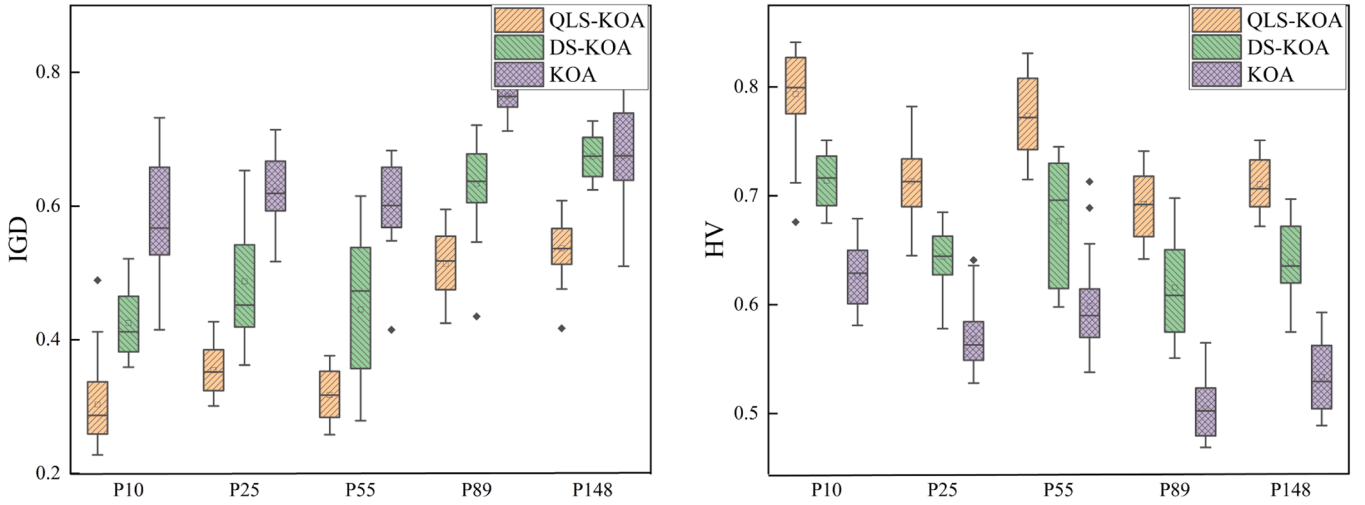


Fig. 7. Q-learning local search framework.

Table 2

The comparison algorithms optimal values across various cases.

Case	UT	QLS-KOA			DS-KOA			KOA		
		f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3
P10	40	22.00	576.87	2.92	22.00	576.87	2.92	22.00	576.87	2.92
	50	23.40	637.91	2.98	23.40	637.91	2.98	23.62	638.15	3.25
	60	22.45	690.72	2.62	22.45	690.72	2.62	22.45	499.10	2.92
	70	27.02	632.75	2.62	28.76	613.82	3.06	34.51	503.16	2.94
	80	31.76	652.72	2.62	34.72	603.34	2.68	45.67	476.34	2.95
P25	30	30.12	1489.25	4.89	33.45	1424.58	4.89	37.75	1423.21	4.96
	40	33.62	1454.87	4.44	33.62	1454.87	4.44	42.36	1448.12	4.51
	50	46.48	1650.55	4.44	48.62	1613.28	4.55	58.72	1615.19	4.6
	60	57.61	1386.32	4.42	63.92	1289.45	4.53	72.68	1268.95	4.45
P55	70	62.01	1527.35	4.38	65.47	1530.21	4.54	76.22	1430.42	4.59
	50	62.01	1533.02	4.38	65.47	1528.19	4.54	76.22	1524.95	4.59
	60	39.31	2992.15	12.83	46.18	2958.47	15.05	49.08	2884.79	15.01
	70	46.92	3356.98	12.94	66.82	3340.12	13.87	73.65	3182.87	14.05
P89	80	45.32	3008.45	13.12	81.32	2979.83	13.39	71.92	2910.89	13.67
	90	57.48	2998.47	12.22	97.45	2995.13	12.94	98.92	2787.42	12.64
	150	180.25	5270.32	48.95	209.68	5065.19	51.22	223.21	5009.98	51.43
	200	258.02	5152.23	48.16	293.21	5109.75	49.66	276.58	5142.13	49.48
P148	250	329.68	4905.03	48.31	396.45	4702.68	49.12	343.67	4858.97	49.33
	300	367.88	5072.61	47.41	408.82	4882.75	47.56	370.42	4777.43	47.8
	350	414.58	5171.75	47.29	481.45	4963.58	47.81	474.52	5067.92	48.4
	300	442.85	8891.53	62.36	582.42	8480.27	65.6	587.72	8195.36	65.72
P148	350	561.52	8900.21	64.21	682.32	8694.65	65.57	621.45	8366.37	67.87
	400	644.02	8916.17	65.33	738.08	8709.25	66.2	711.05	7860.51	65.84
	450	651.85	8405.17	65.09	786.72	8093.92	65.63	732.62	7260.34	65.42
	500	638.18	8869.32	65.21	696.65	8760.19	65.3	658.47	7800.21	65.55

**Fig. 8.** The outcomes of IGD and HV.

the vis-viva equation Eq. (18). Velocity is calculated in two stages: the first enhances solution diversity by determining the velocities of planets near the sun, adjusting them through an upper and lower limit-based step size; the second calculates the velocities of planets farther from the sun to further augment the solution's diversity.

$$V_i(t) = \begin{cases} \left(\vec{X}_a - \vec{X}_i \right) \times r_4 \times H + \mathcal{F} \times \vec{U}_2 \times \vec{r}_5 (1 - R_{i-\text{norm}}(t)) \\ \quad \times \left(r_3 \vec{X}_{i,\text{up}} - \vec{X}_{i,\text{low}} \right), & \text{if } R_{i-\text{norm}}(t) > 0.5 \\ \left(2r_4 \vec{X}_i - \vec{X}_b \right) \vartheta + \left(\vec{X}_a - \vec{X}_b \right) \vartheta^* + \mathcal{F} \times \vec{U}_1 \times \vec{r}_5 \\ \quad \times (1 - R_{i-\text{norm}}(t)) \times \left(\vec{X}_{i,\text{up}} - \vec{X}_{i,\text{low}} \right), & \text{Else} \end{cases} \quad \# \quad (18)$$

Where

$$H = \sqrt{\mu \times (M_s + m_i) \left| \frac{2}{R_i(t) + \varepsilon} - \frac{1}{a_i(t) + \varepsilon} \right|} \quad \# \quad (19)$$

$$\vartheta = \vec{U} \times (r_3 \times (1 - r_4) + r_4) \times H \quad \# \quad (20)$$

$$\vartheta^* = (1 - \vec{U}) \times (r_3 \times (1 - \vec{r}_5) + \vec{r}_5) \times H \quad \# \quad (21)$$

where $V_i(t)$ denotes the velocity of the i -th planet, r_3 and r_4 are two randomly chosen values uniformly distributed between 0 and 1, and $\vec{r}_5$ is a vector of values generated arbitrarily in the range (0,1); μ denotes the universal gravitational constant; $\vec{X}_a$ and $\vec{X}_b$ are the two objects randomly chosen from all the individuals at present; M_s and m_i denote the masses of the optimal solution X_s and the object X_i , respectively; $R_i(t)$ denotes the distance between X_s and X_i at time t ; ε denotes the minimum value used to avoid the error of division by zero; $R_i(t)$ denotes the distance between the optimal solution X_s and the object X_i at time t ; ε

Table 3
The results of the comparison algorithms.

QLS-KOA			KOA			NSGA-II			VNS-NSGA-II			GSA			AMOGA-QL		
Case	t	UT	f ₁	f ₂	f ₃	f ₁	f ₂	f ₃	f ₁	f ₂	f ₃	f ₁	f ₂	f ₃	f ₁	f ₂	f ₃
P10	4.23	40	22.00	576.87	2.92	22	576.87	2.92	25.16	394.21	3.22	22	576.87	2.92	22	435.44	2.92
	4.45	50	23.40	637.91	2.98	23.40	635.40	3.22	26.02	552.66	3.22	23.40	637.91	3.52	23.40	494.06	2.92
	4.79	60	22.45	690.72	2.62	22.45	690.51	2.62	22.45	499.10	2.92	22.45	690.72	2.92	22.45	506.98	2.92
	4.64	70	27.20	613.60	2.62	28.54	611.28	3.04	34.23	500.65	2.92	27.20	613.61	2.92	28.34	417.44	2.92
P25	4.36	80	32.00	600.53	2.62	34.47	600.52	2.65	45.39	474.01	2.92	32.25	600.52	2.92	34.98	419.92	2.92
	5.52	30	30.35	1420.9	4.91	33.19	1420.9	4.95	37.48	1420.88	4.93	33.02	1418.9	4.93	31.83	1415.7	4.93
	5.39	40	33.87	1458.1	4.46	41.00	1439.3	4.53	42.09	1445.23	4.49	36.29	1444.5	4.52	42.05	1452.9	4.57
	5.98	50	46.74	1614.4	4.46	48.38	1610.4	4.52	58.40	1612.36	4.58	51.83	1609.8	4.48	48.30	1592.7	4.48
P55	6.41	60	57.87	1289.1	4.40	63.64	1286.6	4.50	72.37	1266.23	4.43	63.64	1286.6	4.57	62.24	1268.7	4.85
	6.24	70	62.24	1530.1	4.40	65.19	1527.6	4.51	75.94	1527.68	4.57	66.78	1525.1	4.54	65.16	1527.6	4.51
	7.19	50	39.53	2987.6	12.86	45.94	2987.3	15.01	48.83	2887.46	14.98	45.00	2952.8	14.45	44.61	2482.0	13.81
	7.32	60	47.19	3353.5	12.97	66.54	3338.4	13.84	73.37	3185.65	14.02	72.49	3349.4	13.60	71.17	2953.2	13.25
P89	7.65	70	45.55	3005.0	13.15	81.03	2977.1	13.36	71.65	2913.63	13.64	71.56	2989.4	13.45	77.09	2364.3	12.37
	7.56	80	57.74	2995.3	12.25	97.16	2994.4	12.91	98.64	2790.05	12.61	60.71	2994.4	13.06	91.48	2517.1	12.46
	7.27	90	64.19	3427.4	11.71	108.21	3427.4	12.49	112.37	3305.53	12.49	85.27	3430.4	12.28	103.65	3109.6	12.25
	10.89	150	180.76	5267.0	49.00	209.34	5264.5	51.19	222.89	5212.666	51.40	201.89	5267.0	50.20	189.80	5189.9	51.13
P148	10.30	200	258.43	5148.9	48.19	292.88	5148.6	49.63	276.23	5145.69	49.45	295.23	5148.7	48.76	274.98	5072.6	49.30
	10.97	250	330.14	4901.7	48.34	396.07	4901.7	49.09	343.31	4861.65	49.30	406.81	4901.7	49.33	355.08	4862.2	48.10
	10.56	300	368.34	5069.38	47.44	408.47	4981.34	47.53	370.07	4980.15	47.77	405.68	5068.8	47.77	378.77	4981.35	47.77
	10.07	350	415.03	5168.43	47.32	481.08	5162.91	47.78	474.15	5070.57	48.37	481.99	5165.92	48.37	428.97	5165.92	48.37
	15.86	300	443.34	8888.07	62.39	582.05	8879.55	65.57	587.34	8197.97	65.69	567.86	8883.49	65.84	466.66	7803.87	65.48
	15.95	350	562.06	8896.83	64.24	681.95	8893.98	65.54	621.06	8369.26	67.84	645.54	8897.23	66.33	735.88	7897.23	67.52
	15.74	400	644.54	8912.84	65.36	737.71	8908.41	66.17	710.69	7863.19	65.81	740.60	8911.89	65.72	766.38	7654.58	65.51
	15.38	450	652.31	8401.79	65.12	786.34	8392.79	65.60	732.25	7262.96	65.39	753.08	8395.78	65.60	752.25	7091.83	65.24
	15.83	500	638.64	8865.99	65.24	696.27	8859.49	65.27	658.09	7802.76	65.52	699.38	8862.49	65.60	687.13	7314.00	65.51

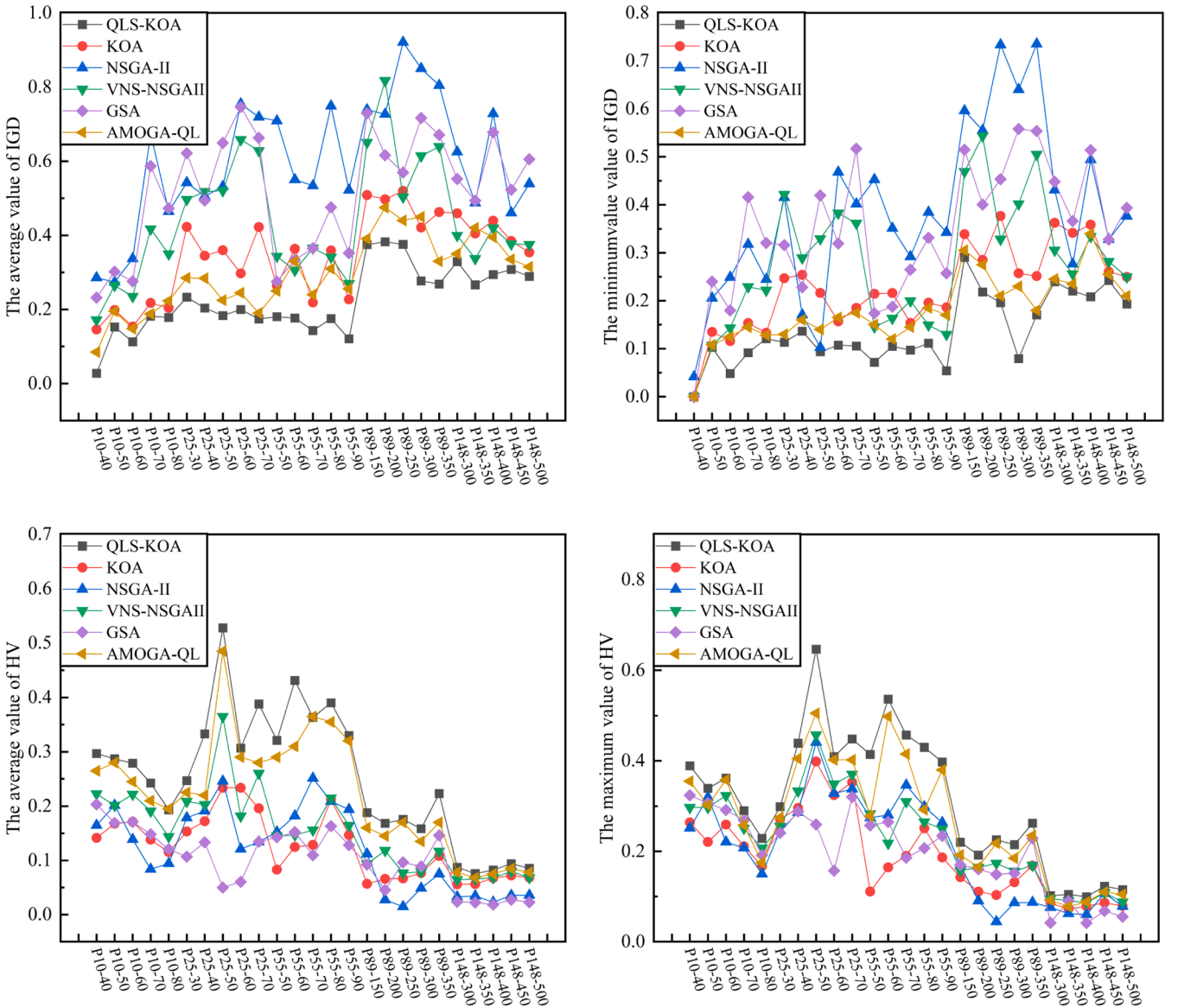


Fig. 9. The outcomes of IGD and HV.

denotes the minimum value used to avoid divide-by-zero error; and $a_i(t)$ is the elliptical orbital half-length axis of the object X_i at time t , defined by Kepler's third law, as shown in Eq. (22).

$$a_i(t) = \left[\mu(t) \times (M_s + m_i) \times \frac{T_i^2}{4\pi^2} \right]^{\frac{1}{3}} \times r_3 \# \quad (22)$$

Where T_i represents the orbital period of object X_i , expressed as an absolute value randomly generated from a normal distribution. In the algorithm, the semi-major axis of the elliptical orbit of object X_i diminishes as the solution approaches the region containing the potential global optimum, and $R_{i-\text{norm}}(t)$ denotes the normalization of the Euclidean distance between X_s and X_i , defined as follows.

$$R_{i-\text{norm}}(t) = (R_i(t) - R_{\min}(t)) / (R_{\max}(t) - R_{\min}(t)) \# \quad (23)$$

4.5. Updating objects' positions

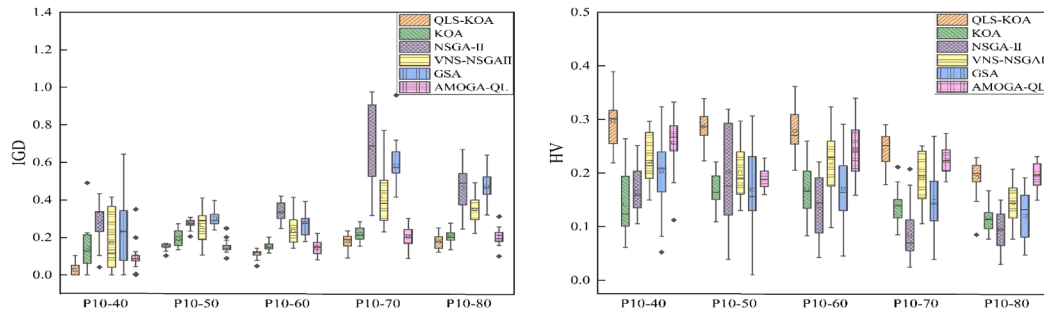
The QLS-KOA simulates the behavior of a celestial body traversing an elliptical orbit around the Sun and seeks the optimal solution through two distinct phases: exploration and exploitation. During the exploration phase, it emphasizes objects situated at considerable distances from

the Sun to uncover new solutions, whereas in the exploitation phase, solutions in proximity to the Sun are meticulously utilized to approach the optimal solution. The QLS-KOA comprehensively addresses the entire search space by updating the position of each object distant from the Sun using the corresponding equation, thereby enhancing search efficiency, as demonstrated in Eq. (24).

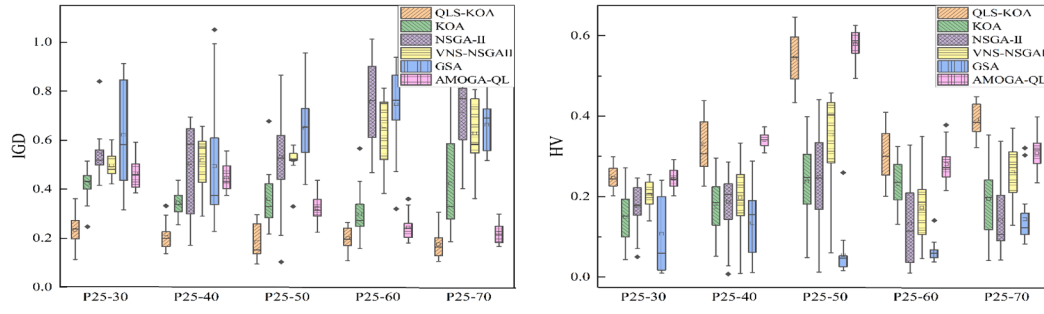
$$\vec{X}_i(t+1) = \vec{X}_i(t) + \mathcal{F} \times \vec{V}_i(t) + \left(\vec{X}_s(t) - \vec{X}_i(t) \right) \times (F_g(t) + |r|) \times \vec{U} \# \quad (24)$$

Where $\vec{X}_i(t+1)$ represents the new position of object i at time $t+1$, $\vec{X}_s(t)$ denotes the optimal position of the Sun identified thus far, and $\vec{V}_i(t)$ signifies the velocity necessary for object i to attain the new position. Additionally, $\mathcal{F}$ serves as the indicator employed to modify the search direction, while $\vec{U}$ constitutes a binary vector. $F_g(t)$ represents the gravitational attraction between the Sun X_s and object X_i , calculated according to the law of gravitation, as illustrated in Eq. (25).

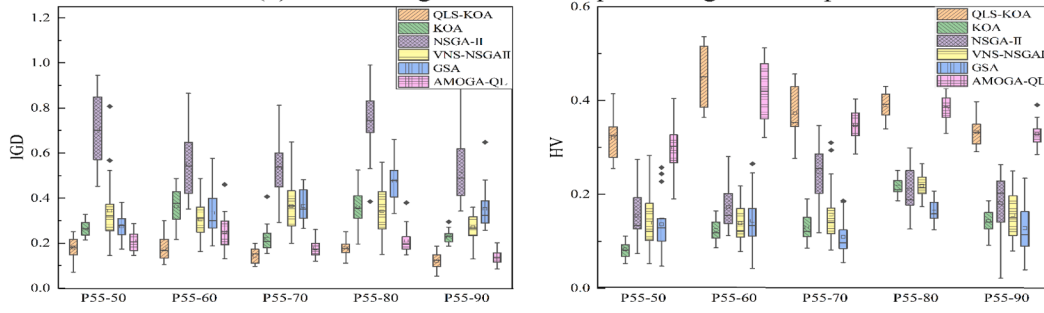
$$F_g(t) = r_4 + e_i \times \mu(t) \times (\overline{M}_s \times \overline{m}_i) / (\overline{R}_i^2 + \varepsilon) \# \quad (25)$$



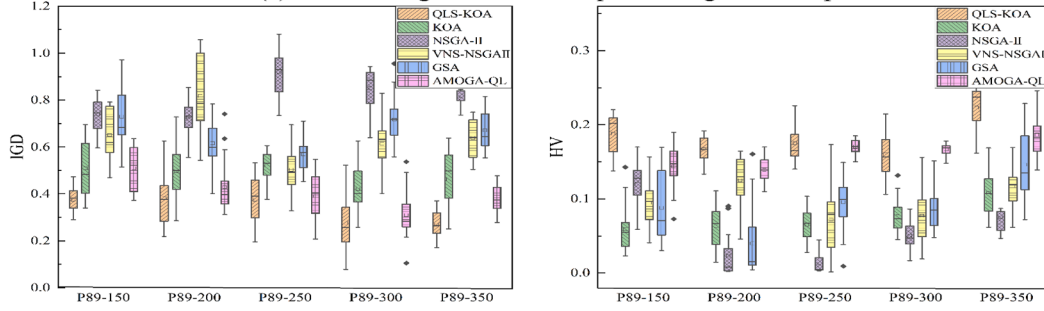
(a) The box diagram of the comparison algorithm in p10.



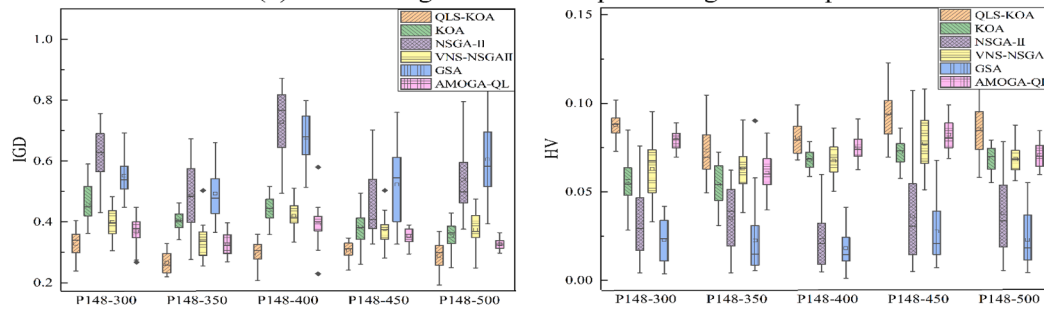
(b) The box diagram of the comparison algorithm in p25.



(c) The box diagram of the comparison algorithm in p55.



(d) The box diagram of the comparison algorithm in p89.



(e) The box diagram of the comparison algorithm in p148.

Fig. 10. The box plots of algorithms performance across five cases.

Table 4

The results of statistics analysis on IGD.

Case	QLS-KOA			KOA		NSGA-II		VNS-NSGA-II		GSA		AMOGA-QL	
	UT	WT	FT	WT	FT	WT	FT	WT	FT	WT	FT	WT	FT
P10	40	-	1.525	2.30e-04	2.625	1.078e-07	4.200	1.728e-04	3.000	3.904e-05	3.650	8.52E-06	1.850
	50	-	1.500	6.863e-04	2.050	6.031e-08	3.725	2.355e-04	3.300	6.241e-08	4.425	5.17E-07	1.780
	60	-	1.050	1.626e-06	2.100	6.403e-08	4.550	6.532e-08	3.300	6.523e-08	4.000	1.23E-06	1.655
	70	-	1.350	2.070e-02	1.700	6.578e-08	4.400	1.235e-07	3.300	1.235e-07	4.250	9.86E-05	1.720
	80	-	1.300	3.600e-02	1.700	9.0977e-08	4.300	1.657e-07	3.250	6.738e-08	4.450	1.52E-04	1.850
P25	30	-	1.050	3.411e-07	2.550	6.786e-08	3.950	6.757e-08	3.475	1.059e-07	3.975	4.69E-07	1.755
	40	-	1.150	5.227e-07	2.750	1.592e-05	3.700	1.065e-07	3.800	2.956e-07	3.600	8.34E-06	1.820
	50	-	1.050	2.062e-06	2.450	2.356e-06	3.500	6.588e-08	3.525	6.767e-08	4.475	3.12E-06	1.680
	60	-	1.150	4.680e-05	1.850	6.786e-08	4.050	6.700e-08	3.700	6.796e-08	4.250	2.75E-05	1.79
	70	-	1.050	1.376e-06	2.450	6.691e-08	4.425	6.757e-08	3.500	6.719e-08	3.575	7.94E-07	1.715
P55	50	-	1.200	3.499e-06	2.550	6.796e-08	4.950	2.166e-05	3.300	1.600e-05	3.000	5.28E-06	1.850
	60	-	1.000	3.416e-07	3.450	6.795e-08	4.750	8.589e-06	2.750	2.353e-06	3.050	9.63E-07	1.765
	70	-	1.0750	1.174e-05	2.025	6.796e-08	4.650	7.898e-08	3.700	6.796e-08	3.550	6.82E-06	1.810
	80	-	1.150	1.655e-07	2.800	6.786e-08	4.900	7.569e-06	2.400	6.786e-08	3.750	4.19E-07	1.735
	90	-	1.000	7.306e-08	2.250	6.796e-08	4.800	2.960e-07	2.900	6.796e-08	4.050	3.57E-08	1.695
P89	150	-	1.150	2.221e-04	2.200	6.796e-08	4.100	7.898e-08	3.350	6.796e-08	4.200	1.38E-04	1.885
	200	-	1.400	7.100e-03	2.050	9.160e-08	4.150	1.655e-07	4.500	4.535e-06	2.900	5.27E-04	1.925
	250	-	1.325	1.807e-05	2.800	6.786e-08	5.000	8.768e-04	2.625	1.047e-06	3.250	1.13E-05	1.870
	300	-	1.250	1.794e-04	1.900	6.748e-08	4.650	2.218e-07	3.200	6.796e-08	4.000	1.09E-04	1.830
	350	-	1.050	9.1277e-07	2.100	6.403e-08	4.900	6.796e-08	3.400	6.796e-08	3.550	5.68E-07	1.740
P148	300	-	1.100	3.939e-07	3.000	6.796e-08	4.650	1.0373e-04	2.150	6.796e-08	4.100	2.45E-07	1.945
	350	-	1.200	6.786e-08	3.350	1.655e-07	4.150	3.922e-05	2.000	6.786e-08	4.300	2.18E-05	1.865
	400	-	1.000	9.1730e-08	2.700	6.796e-08	4.650	1.918e-07	2.300	6.796e-08	4.350	5.09E-08	1.775
	450	-	1.200	1.699e-05	2.950	2.381e-07	3.750	2.041e-05	2.900	2.059e-07	4.200	9.83E-06	1.890
	500	-	1.400	2.7451e-04	2.150	6.796e-08	4.250	2.919e-05	2.600	6.796e-08	4.600	1.67E-04	1.955

Table 5

The results of statistics analysis on HV.

Case	QLS-KOA			KOA		NSGA-II		VNS-NSGA-II		GSA		AMOGA-QL	
	UT	WT	FT	WT	FT	WT	FT	WT	FT	WT	FT	WT	FT
P10	40	-	4.850	1.657e-07	1.750	2.487e-05	1.950	4.680e-05	3.300	6.558e-05	3.150	3.25E-05	4.215
	50	-	4.500	6.796e-08	2.300	1.220e-02	2.050	1.807e-05	3.025	1.804e-05	2.125	9.68E-06	4.050
	60	-	4.700	6.917e-07	2.400	1.052e-07	1.900	2.300e-03	3.500	3.037e-06	2.500	1.82E-06	4.350
	70	-	4.700	1.657e-07	2.500	1.197e-07	1.450	8.357e-04	3.650	4.494e-06	2.700	2.98E-06	4.415
	80	-	4.650	1.374e-06	2.500	4.499e-07	1.800	1.792e-04	3.400	3.470e-06	2.650	2.15E-06	4.150
P25	30	-	4.550	2.356e-06	2.400	3.288e-05	2.950	6.220e-04	3.300	8.033e-06	1.800	5.17E-06	4.230
	40	-	4.800	2.062e-06	2.700	1.574e-06	2.600	1.250e-05	3.050	2.562e-07	1.850	8.34E-07	4.315
	50	-	5.000	6.796e-08	2.550	7.898e-08	2.650	3.416e-07	3.650	6.767e-08	1.150	8.95E-08	4.330
	60	-	4.600	3.048e-04	3.600	4.535e-06	2.300	7.562e-06	3.100	6.776e-08	1.400	2.17E-06	4.145
	70	-	5.000	1.065e-07	2.900	7.734e-08	1.750	3.411e-07	3.500	6.776e-08	1.850	9.26E-08	4.450
P55	50	-	5.000	6.776e-08	1.450	1.427e-07	3.250	1.427e-07	2.700	1.062e-07	2.600	1.25E-07	4.540
	60	-	4.850	6.796e-08	1.950	6.796e-08	3.200	6.796e-08	2.400	6.786e-08	2.450	7.93E-08	4.250
	70	-	5.000	6.786e-08	2.050	1.046e-06	3.800	1.429e-07	2.600	6.786e-08	1.550	9.86E-08	4.650
	80	-	5.000	6.796e-08	3.050	6.796e-08	2.400	6.796e-08	3.200	6.796e-08	1.350	8.54E-08	4.350
	90	-	5.000	6.796e-08	2.350	6.796e-08	3.300	6.796e-08	2.450	6.796e-08	1.900	9.12E-08	4.450
P89	150	-	4.950	9.173e-08	1.650	5.227e-07	3.450	1.918e-07	2.550	7.948e-07	2.400	3.18E-07	4.360
	200	-	4.950	6.786e-08	2.700	6.748e-08	1.600	2.039e-05	3.650	1.918e-07	2.100	1.25E-06	4.450
	250	-	4.900	6.796e-08	2.550	6.709e-08	1.150	6.917e-07	2.750	9.173e-08	3.650	4.82E-08	4.270
	300	-	4.800	1.235e-07	2.700	6.467e-08	1.400	6.015e-07	2.900	6.917e-07	3.200	8.73E-08	4.170
	350	-	4.950	1.655e-07	2.400	6.243e-08	1.300	1.4311e-07	2.850	8.597e-06	3.500	1.09E-07	4.350
P148	300	-	4.900	3.934e-07	3.250	7.876e-08	1.950	2.353e-06	3.550	6.786e-08	1.350	1.85E-07	4.440
	350	-	4.550	2.745e-04	3.100	5.227e-07	2.000	4.8e-02	3.700	1.120e-06	1.650	1.85E-06	4.050
	400	-	4.700	6.239e-05	3.650	6.786e-08	1.550	3.749e-04	3.600	6.796e-08	1.500	3.12E-05	4.255
	450	-	4.600	7.577e-06	3.400	9.127e-07	1.850	3.966e-03	3.650	6.796e-08	1.500	4.28E-06	4.155
	500	-	4.700	3.042e-04	3.550	9.766e-07	1.900	2.7451e-04	3.500	6.796e-08	1.350	1.89E-05	4.345

Where e_i denotes the eccentricity of the planetary orbit, its value ranges from 0 to 1, thereby introducing a degree of randomness into the algorithm's solution. Concurrently, the normalized values of the planetary mass $\bar{M}_s$ and the solar mass $\bar{m}_i$ are defined by the mass indices $\bar{M}_s$ and $\bar{m}_i$. The equations formulated are utilized to model the gravitational relationship between the Sun and the planets, which constitutes a pivotal factor in optimizing planetary motions and orbits. The standardized values $\bar{M}_s$ and $\bar{m}_i$ can be represented by the Eq. (26) and Eq. (27).

$$\bar{M}_s = r_2 \times (f_s(t) - \text{worst}(t)) / \sum_{k=1}^N (f_k(t) - \text{worst}(t)) \# \quad (26)$$

$$\bar{m}_i = (f_i(t) - \text{worst}(t)) / \sum_{k=1}^N (f_k(t) - \text{worst}(t)) \# \quad (27)$$

Where $f_k(t)$ represents the fitness value for each position of individual k at the current time t ; $f_s(t)$ indicates the current fitness value of the optimal individual, referred to as the Sun and $\text{worst}(t)$ signifies the fitness value of the worst solution candidate.

$\bar{R}_i$ represents the normalized value of R_i , signifying the Euclidean distance between individual X_s and individual X_i , as expressed by Eq. (28).

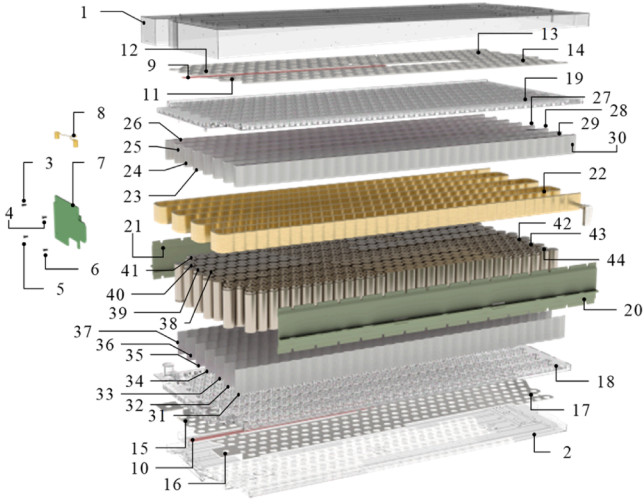


Fig. 11. Tesla Model S battery module.

$$R_i(t) = \|X_S(t) - X_i(t)\|_2 = \sqrt{\sum_{j=1}^{\dim} (X_{Sj}(t) - X_{ij}(t))^2} \# \quad (28)$$

$\mu(t)$ as a function of control search refinement, exhibits an exponential decrease over time t , as illustrated in Eq. (29).

$$\mu(t) = \mu_o \times e^{-\gamma \left(\frac{t}{MAXGEN} \right)} \# \quad (29)$$

Where μ_o represents the initial value, γ is a constant, and t and $MAXGEN$ denote the current iteration number and the maximum iteration number, respectively.

4.6. Q-learning-based local search

In order to accomplish a thorough exploration of the disassembly part order vector and the disassembly operation decision vector, a dynamic proportional local search framework based on Q-learning is proposed, as depicted in Fig. 7. Within this framework, Q-learning determines the suitable set of actions according to the current solution state, realizing dynamic adjustments to the proportions of the four domain search operators. As environmental rewards are conveyed to the intelligent agent, the Q-table undergoes iterative updates, culminating in the derivation of the optimal strategy. The Q-table update rule is as follows:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)) \# \quad (30)$$

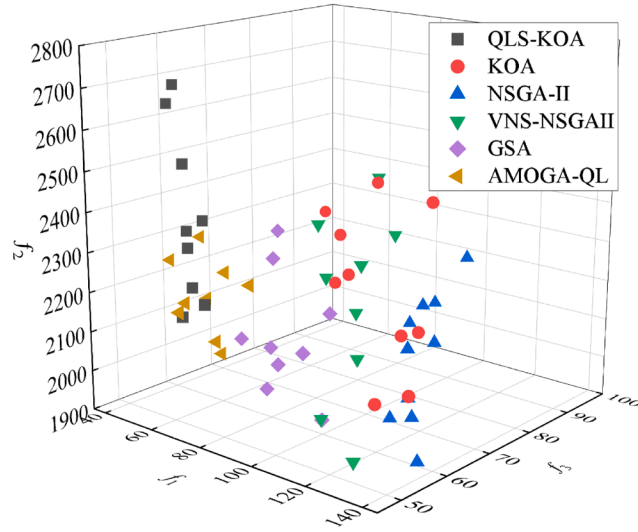
Where α is the learning rate, γ indicates the discount factor, r_{t+1} is the reward obtained from the environment by taking an action a_t in the state s_t .

4.6.1. State set

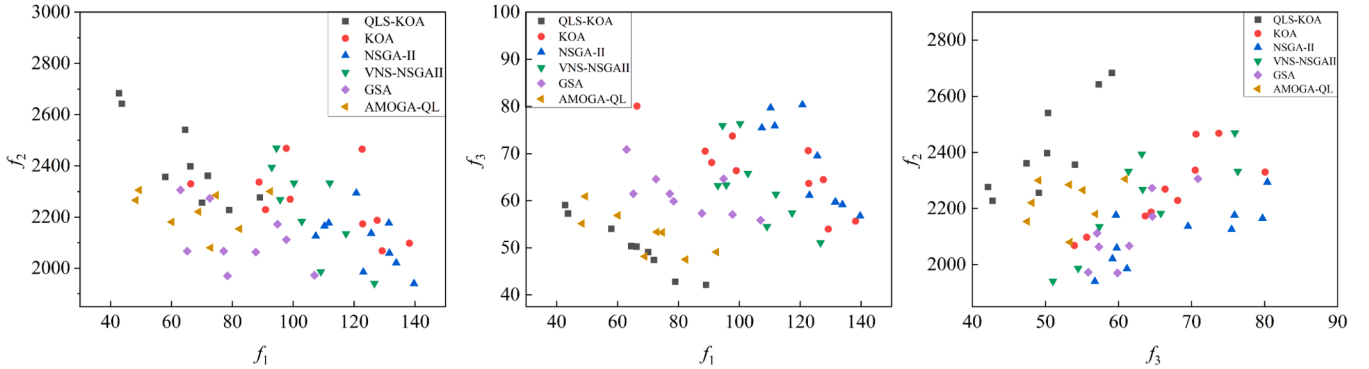
This study adopts the Inverted Generational Distance (IGD) [20] as the fitness metric to assess the quality of individuals. The formation of the environmental state should prioritize the mean fitness across the population, its diversity, and the excellence of individual fitness. Eq. (31) outlines the method for normalizing the average fitness value in generation t . Eq. (32) outlines the method for normalizing the genetic diversity in generation t . Eq. (33) formulates the approach for normalizing the optimal fitness value. Eq. (34) represents the population state value.

Table 6
The optimum outcomes attained by six algorithms.

t	UT	QLS-KOA			KOA			NSGA-II			VNS-NSGAI			GSA			AMOGA-QL		
		f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3
8.64	200	36.37	2209.42	45.42	42.41	2192.22	46.32	54.65	2205.25	49.26	43.89	2206.70	45.93	39.67	2200.39	46.77	40.15	2258.17	46.77
8.54	300	104.58	2649.18	45.42	138.59	2644.78	47.42	166.41	2079.67	47.37	116.07	2648.69	47.42	113.96	2358.41	46.32	109.25	2648.93	45.87
8.80	400	233.41	2652.48	47.57	250.23	2646.43	48.92	297.02	2094.68	49.37	271.15	2649.43	47.87	280.82	2303.65	47.57	241.85	2651.27	47.72



(a) The NDSs obtained by five comparison algorithms.



(b) Distribution of NDSs obtained by comparison algorithms on pairwise objectives.

Fig. 12. The NDSs obtained by the five comparison algorithms.

$$f^* = \frac{\sum_{i=1}^N f(x_i^t)}{\sum_{i=1}^N f(x_i^1)} \# \quad (31)$$

$$d^* = \frac{\sum_{i=1}^N \left| f(x_i^t) - \frac{\sum_{i=1}^N f(x_i^t)}{N} \right|}{\sum_{j=1}^N \left| f(x_j^1) - \frac{\sum_{j=1}^N f(x_j^1)}{N} \right|} \# \quad (32)$$

$$m^* = \frac{\max f(x_i^t)}{\max f(x_i^1)} \# \quad (33)$$

$$S^* = w_1 f^* + w_2 d^* + w_3 m^* \# \quad (34)$$

The fitness value of the i -th member in the t -th iteration is denoted by $f(x_i^t)$, N is the size of the group, w_1 , w_2 , w_3 are the weight values and sum to 1. Population average fitness and diversity contribute to overall quality, fostering the emergence of superior individuals, thus w_1 , w_2 , w_3

take values of 0.35, 0.35, and 0.3, in that order.

An excess or deficiency of states can exert an impact on both convergence and the quality of outcomes in algorithm. So, it necessitates a delicate balance with exploration costs. In this paper, the algorithm's state range, denoted as S^* , spans from 0 to 1, exclusively. This set of states is partitioned into 20 discrete intervals: $[s(1), s(2), \dots, s(19), s(20)]$, each interval having a width of 0.05 [23]. For instance, when S^* falls within the range of $[0, 0.05]$, s belongs to $s(1)$; when S^* falls within $[0.05, 0.10]$, s belongs to $s(2)$, and so forth.

4.6.2. Dynamic proportion search action set

In order to improve the search efficiency of the algorithm, this study dynamically selects four different neighborhood structure generation methods designed for HRC-PDLBP as the selection of action states through the agent, as shown in Fig. 7. These encompass a crossover reordering operator, a mutation reordering operator, a bilateral insertion operator, and a DV reconstruction operator. In each population generation, agents undertake diverse actions to select the appropriate neighborhood search ratio ($S_r = \{r_t \in (0, 1) \mid t = 1, 2, 3, 4\}$, $r = 1, 2, \dots, 9, 10$), which consist of 10 possible actions. Here, r_t consists of four decimals totaling 1, delineating the proportions of four distinct neighborhood searches within the populations. The detailed descriptions of the neighborhood structure generation methods are articulated as

Table 7

Task schemes of induction Tesla Model S battery module.

Solution	f_1	f_2	f_3	Task scheme
1	39.67	2130.05	45.87	CW1:1→2→3→6→5→4→7→9; CW2:11→13; CW3:12→14; CW4:10→18→21→8; CW5:15→16; CW6:17; CW7: 19→20→22→26→34; CW8:29→36→27→28→30→ 23 → 31→25→24→37; CW9:38; CW10:44.
2	36.37	2130.36	45.87	CW1:2→1→3→4→5→6→7→9; CW2:11→12; CW3:14→13; CW4:18→21→20→10; CW5:15→16; CW6: 22→27→31→23; CW7:38; CW8:17; CW9:24→30→19→8→29 →28 →36→34→37→26→34→ 37→26→ 25; CW10:44.
3	39.99	2209.42	49.77	CW1:1→2→3→6→5→4→7→9; CW2:11→13; CW3:12→14; CW4:10→18→21→8; CW5:15→16; CW6:17; CW7: 19→20→22→26 →34; CW8: 29→36→27→28→30→ 23→31→25→24→37; CW9:42; CW10:44; CW11:38.
4	125.69	2069.75	45.42	CW1: 1→2→3→4→5→6→7→8; CW2:9→10; CW3:11→12; CW4:13→14; CW5:15→ 16; CW6:17; CW7:18→19→20→21; CW8:22→23; CW9: 24→25→26→27→28→29→ 30→ 31→36→37; CW10:38; CW11:44.

follows.

- (1) The procedures for the crossover reordering operator are delineated below:

Step 1: Acquire the TV vector and arbitrarily designate two crossover points; the segment between these points constitutes the crossover fragment, denoted as {9, 6, 4, 7}.

Step 2: Transfer the genetic material not within the crossover fragment of Parent 1 (P_1) into Child 1 (C_1), maintaining their original locations within P_1 .

Step 3: Replicate all genes of the crossover fragment from Parent 2 (P_2) to Child 1 (C_1) in the sequence they appear in P_2 .

Step 4: Inherit the DV vector from Parent 1.

- (2) The steps for the mutation reordering operator are outlined as follows:

Step 1: Attain the TV vector and randomly designate two mutation points; the segment between these points forms the mutation fragment, identified as {4, 6, 7, 8, 9}.

Step 2: Revise the priority matrix, retaining only the task priority relations within the mutation fragment.

Step 3: Produce a fresh disassembly sequence derived from the amended matrix.

Step 4: Inherit the DV vector from the parent.

- (3) The procedures for the bilateral insertion operator are outlined as follows:

Step 1: Acquire the TV vector and randomly designate a mutation point, identified as {6}.

Step 2: Retrieve the task set immediately following the mutation point and the subsequent task set, denoted as {9, 4} and {10}. It is evident that the range for randomly mutating positions lies between tasks 9 and 10.

Step 3: Randomly insert the mutation point task into the sequence of constrained tasks.

Step 4: Inherit the DV vector from the parent.

- (4) The steps for the DV reconstruction operator are as follows:

Step 1: Attain the DV vector.

Step 2: Reconstruct the DV in accordance with the priority matrix.

4.6.3. Reward method

The reward stands as a pivotal metric for assessing the actions chosen by the agent, accumulating rewards through repeated action selection to eventually deduce the optimal outcome. Reward values are influenced by both the average fitness and the best fitness. Regarding S^* , if the present average fitness falls below the previous state, the reward value is positive; otherwise, it is negative. The modification of the reward function is articulated as follows:

$$r = r_1 + r_2 \# \quad (35)$$

$$r_1 = \frac{\sum_{i=1}^N f(x_i^t) - \sum_{i=1}^N f(x_i^{t-1})}{\sum_{i=1}^N f(x_i^{t-1})} \# \quad (36)$$

$$r_2 = \frac{\max f(x_i^t) - \max f(x_i^{t-1})}{\max f(x_i^{t-1})} \# \quad (37)$$

4.6.4. Improved ε -greedy strategy

Defined by Eq. (38), the ε -greedy strategy balances between exploring new options and exploiting known ones in selecting actions. Here, ε , symbolizing the rate of exploitation, and r_{0-1} , an arbitrary value ranging from 0 to 1, play crucial roles. If ε is less than or equal to r_{0-1} , the system follows the greedy strategy, selecting the action 'a' maximizing the anticipated Q-value. Conversely, if ε exceeds r_{0-1} , exploration occurs, and an arbitrary action 'a' is chosen.

$$\pi(s_t, a_t) = \begin{cases} \max Q(s_t, a_t) & r_{0-1} \geq \varepsilon \\ \text{Randomly} & r_{0-1} < \varepsilon \end{cases} \quad (38)$$

In the traditional ε -greedy strategy, there exists a balance issue between exploration and exploitation [25]. To address this, a dynamic adjustment factor, denoted as ρ , is introduced to modulate the value of ε . In the algorithm's initial phases, ε is maximized to promote the exploration of novel solution spaces by the agent. With each iteration of the algorithm, the value of ρ progressively increases, leading to a gradual reduction in ε . This adaptive tuning steer the agent towards favoring actions associated with higher Q values, all while preserving acquired knowledge. The incorporation of the balancing factor effectively harmonizes the delicate interplay between exploration and exploitation, thereby enhancing the efficiency and efficacy of the Q-learning algorithm.

$$\varepsilon = 0.9 - 0.85 \times \rho, \rho = \frac{[t/\text{Maxgen}]}{10} \# \quad (39)$$

5. Numerical experiment and comparison

This section demonstrates the algorithm's superiority through three experiments. The initial experiment compares QLS-KOA with the original KOA and three other renowned algorithms based on the achieved objective optima. Results indicate that QLS-KOA excels in solving HRC-PDLBP. The subsequent experiment further validates the superiority of QLS-KOA using two evaluation metrics, IGD and HV. For a more comprehensive analysis, the final experiment underwent statistical analysis using the 5 % exact Wilcoxon rank sum test and the non-parametric Friedman test to verify the statistical significance of QLS-KOA's results. All trials took place on a computer housing an Intel(R) Core (TM) i5 CPU, 2.50 GHz, and 16.00 GB of RAM.

5.1. Parameters settings

Since this paper focuses its research on HRC-PDLBP for the first time, there is no corresponding benchmark dataset. Therefore, this paper selects five cases from the existing literature and extends them into 25 test datasets based on different cycle times. These consist of sets of 10

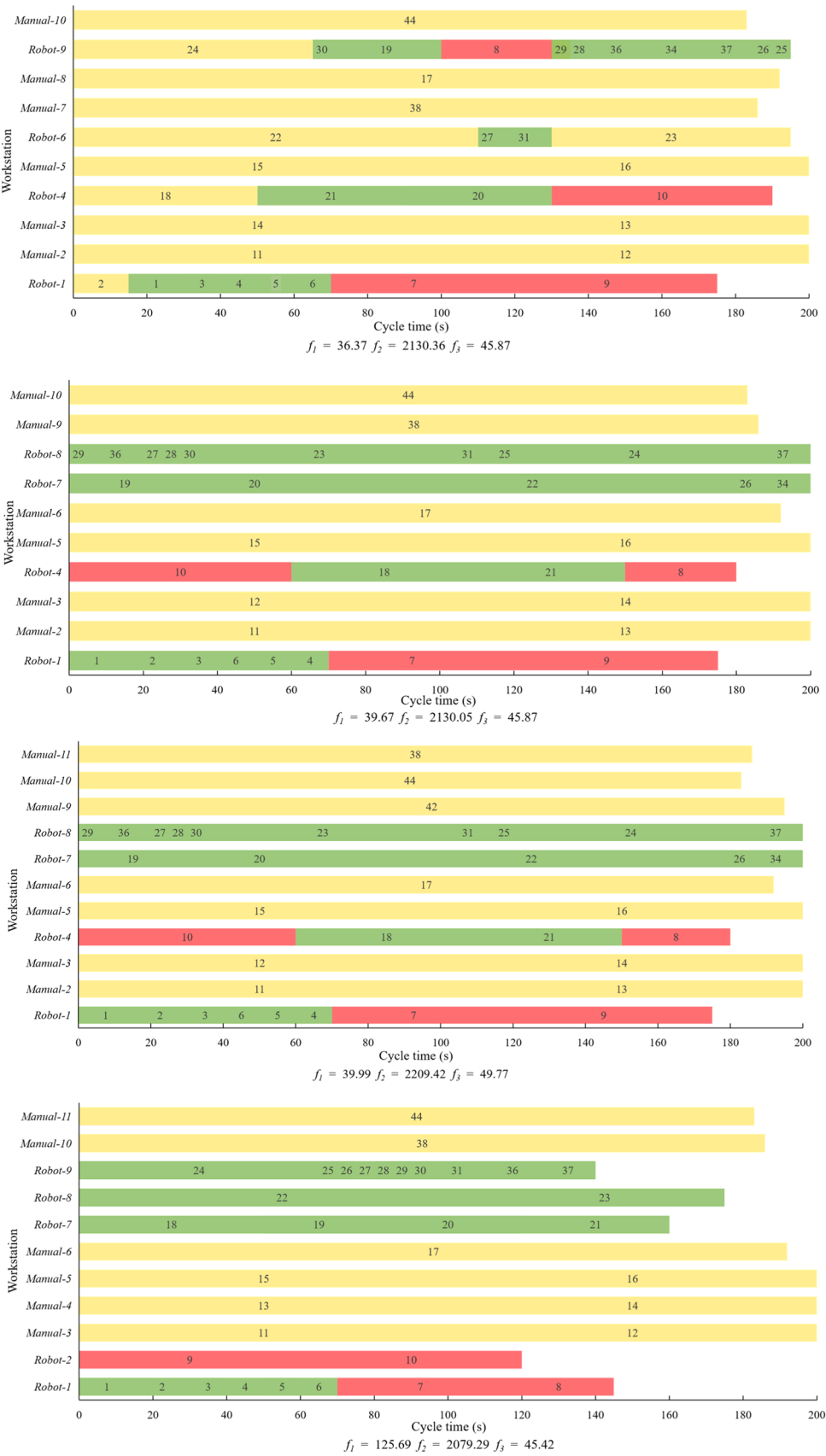


Fig. 13. Task schemes diagram.

components [19], 25 components from a Samsung SCH-3500 mobile phone [16], 55 components from a printer [29], 89 components [7] from one sample and 148 components [3]. The model incorporates specific parameter values, assumed as follows: C_w equals 0.5, C_r equals 2, C_{pt} equals 6/3600, C_{rt} equals 3/3600, C_h equals 0.01, E_{pt} equals 0.02, E_{rt} equals 0.03, and E_h equals 0.01. Additionally, the disassembly profit for each disassembly task is determined as a random number within the range of 20–100.

Three comparison algorithms, namely NSGA-II [34], VNS-NSGAI [26], GSA [31], and AMOGA-QL [33] are employed to illustrate the superior performance of QLS-KOA across various scales. Parameter configurations for each algorithm are determined through orthogonal experiments. The parameter settings for QLS-KOA are as follows: $Popnum$ is 50, $Maxgen$ is 50, r is 10, μ_0 is 0.2, $\bar{T}$ is 4, α is 0.5 and γ is 0.4. Due to the distinct structures of each algorithm, employing an identical iteration count as a termination criterion is not reasonable. Thus, within this manuscript, the mean time (t) from 20 runs for QLS-KOA serves as the stopping criterion for the five algorithms. The $Popnum$, r , μ_0 , and $\bar{T}$ of KOA are set to the same values as for QLS-KOA. The $Popnum$, P_c and P_m of NSGA-II are set to 50, 0.8, and 0.25, respectively. VNS-NSGAI's $Popnum$, P_c and P_m are set to 50, 0.8, and 0.1, respectively. The $Popnum$, P_c , P_m , T_s , T_c and γ of GSA are set to 50, 0.8, 0.2, 200, 10, and 0.974, respectively. The $Popnum$, P_c , P_m , α and γ of AMOGA-QL are set to 50, 0.8, 0.2, 0.9 and 0.1. Additionally, Inverted Generational Distance (IGD) and Hypervolume (HV) [26] are employed to assess the performance of the comparative algorithms.

5.2. Ablation experiments

To validate the effectiveness of the Q-learning-based dynamic operator proportion adjustment mechanism, ablation experiments are conducted on five benchmark instances using three algorithms: Original KOA (without local search), DS-KOA (with fixed operator proportions), and QLS-KOA. The evaluation employed IGD, HV, and three core objective values. Table 2 indicates that QLS-KOA, the original KOA, and DS-KOA are all capable of reaching the optimal solution in small cases. However, in complex scenarios, only QLS-KOA achieves load balancing and energy consumption values closer to the coordinate origin, with disassembly profit values concentrated in the high-value region. These findings substantiate that, through the integration of the Q-learning mechanism, QLS-KOA adaptively allocates search resources to explore a broader solution space and effectively circumvent local optima. This fundamental advantage, absent in both the original KOA and DS-KOA, directly validates the performance enhancement conferred by the Q-learning-based dynamic adjustment strategy.

As illustrated in Fig. 8, QLS-KOA consistently attains the lowest and most stable IGD values across all instances, while simultaneously achieving the highest HV values, thereby clearly demonstrating its superiority in solution accuracy and distribution uniformity.

5.3. Performance comparison and discussion

Table 3 presents the top results achieved by the five optimization algorithms for every criterion across the five distinct UT scenarios. It is evident that the newly proposed QLS-KOA algorithm achieves the best objective results on all test problems, indicating its outstanding performance in solving HRC PDLBP problems of different scales. Additionally, while all five optimization algorithms perform well on modestly-sized problems, the QLS-KOA algorithm demonstrates a significant advantage as the problem size increases.

For a thorough evaluation of the effectiveness of the comparative algorithms, Fig. 9 depicts the Inverted Generational Distance (IGD) and Hypervolume (HV) outcomes over 20 iterations. The experimental findings reveal the superiority of the QLS-KOA algorithm over other algorithms in both IGD and HV metrics across handling cases of various

scales. This underscores the superior convergence and distribution performance of the QLS-KOA algorithm. Fig. 10 illustrates the box plots depicting the performance of the data. The proposed QLS-KOA excels in terms of IGD and HV, displaying optimal and consistent performance across problems of varying magnitudes.

In summary, the QLS-KOA algorithm demonstrates superior performance compared to four comparison algorithms used in this study for addressing the HRC-PDLBP. Furthermore, its effectiveness remains consistent across different problem sizes, highlighting the robustness and excellence of the developed QLS-KOA. This is attributed to the algorithm's unique strategy of independently searching for TV and DV genes in different directions, coupled with thorough exploration of each neighborhood, leading to enhanced search capabilities.

5.4. Statistics analysis

Within this segment, the statistical significance of the devised QLS-KOA was confirmed using a pair of nonparametric analyses: the Wilcoxon test [14] and the Friedman test [24]. The Wilcoxon test (WT) and Friedman test (FT) utilized two commonly acknowledged composite evaluation metrics, HV and IGD. Tables 4 and 5 display the outcomes of the statistical analysis conducted on IGD and HV, respectively.

Based on the outcomes delineated in Tables 4 and 5, it is manifest that QLS-KOA consistently attained the best Friedman means in contrast to other algorithms across diverse scenarios. This indicates that the devised QLS-KOA outperforms the five algorithms subjected to experimentation. The findings of the Wilcoxon test reveal a statistically significant difference between the proposed QLS-KOA and the benchmark algorithms for problems of varying scales. In summary, the proposed QLS-KOA can efficaciously address HRC PDLBP with diverse scales.

6. Case study

In this section, the validity and reliability of the proposed models and methods are scrutinized extensively, employing the Tesla Model S battery module for analysis. This module contains 44 disassembly tasks, including 4 hazardous attribute tasks, 14 complex attribute tasks, 4 high-demand attribute tasks, and 22 ordinary attribute tasks. Fig. 11 illustrates a detailed representation of the structural composition of the Tesla Model S battery module [34].

6.1. Performance comparison

The QLS-KOA methodology has been employed to optimize the disassembly of Tesla Model S battery modules, addressing a case involving 44 distinct tasks with varying cycle time constraints, aimed at resolving the HRC-PDLBP challenge. Table 6 presents the optimal values attained for each objective by the six comparative algorithms, with the superior values highlighted in bold.

To elucidate the distribution of acquired solutions, considering an anticipated cycle time constraint set at 200, Fig. 12(a) presents the non-dominated solution sets (NDSs) generated by five algorithms in 20 iterations for the P44 case. Simultaneously, Fig. 12(b) illustrates the distribution relationships among each pair of objective values for the P44 case. Fig. 12 reveals that, in terms of solution sets, QLS-KOA demonstrates superior convergence compared to the other four algorithms. The solution derived from the QLS-KOA algorithm surpasses those achieved with other algorithms, signifying that the non-dominated solution set produced by QLS-KOA is in closer proximity to the true Pareto front.

6.2. Analysis and discussion of disassembly scheme

The QLS-KOA algorithm, as proposed, ensures not only solution diversity but also the efficacy of the obtained solutions. This enables companies to flexibly select diverse task schedules according to their

specific requirements. Table 7 illustrates, under the scenario with an anticipated cycle time constraint of 200, that QLS-KOA yields four non-dominated solutions. Companies can freely opt for task schedules aligned with their decision-making criteria. For instance, if optimizing profit is the priority, there is an inclination towards selecting Scheme 3. However, if the goal is optimal load balancing, option 2 is recommended. To comprehensively showcase the model's performance, the solutions listed in Table 7 are visualized, as depicted in Fig. 13. Intuitively, Fig. 13 reveals that hazardous attribute tasks (7, 8, 9, 10) are allocated to the robot workstation, while complex attribute tasks (11, 12, 13, 14, 15, 16, 17, 38, 39, 40, 41, 42, 43, 44) are assigned to the human workstation. Additionally, the Demand attribute tasks (7, 8, 38, 44) and the Hazard attribute tasks have been disassembled. Considering Scheme 2 and Scheme 3, for example, in Scheme 2, the total number of decomposition tasks is 36, whereas in Scheme 3, it is 37. The additional task, a complex task 42, necessitates the creation of a new human workstation for disassembly. Consequently, Scheme 2 has fewer workstations than Scheme 3, but Scheme 3 exhibits higher profits and energy consumption than Scheme 2. Similarly, when comparing Scheme 2 and Scheme 4, in Scheme 4, the total number of disassembly tasks is 35, resulting in lower energy consumption compared to Scheme 2 due to the reduction of one task. This also underscores the significance of considering partial disassembly in the model presented in this paper.

In numerical experiments, QLS-KOA showcased optimal performance in addressing HRC-PDLBP industrial practices across diverse scenarios. The magnitude of the expected cycle time constraint, a pivotal parameter, profoundly influences the outcomes. As a key parameter, the magnitude of the expected cycle time constraint has a profound effect on the results. As the time constraint increases, the optimal values of f_1 and f_2 tend to increase and f_3 tends to decrease. Moreover, solutions derived by the QLS-KOA algorithm significantly outclass those obtained by other comparative algorithms. This implies that the non-dominated solution set generated by the QLS-KOA algorithm aligns more closely with the authentic Pareto front.

Building upon these research findings, HRC-PDLBP and QLS-KOA provide actionable frameworks for managerial decision-making in remanufacturing and recycling industries. By incorporating task-specific attributes alongside the processing capabilities and quantities of machines and workers, non-dominated solutions can be derived through priority-driven strategies. The optimal solution can then be selected across three dimensions: load balancing, economic efficiency, and environmental sustainability. The model demonstrates particular suitability for mixed-task environments requiring human-machine collaboration and, through its multi-objective structure, offers significant practical value in aligning economic performance, environmental responsibility, and safety assurance.

7. Conclusion

In this paper, a comprehensive HRC-PDLBP model is introduced, encompassing variables such as workload smoothness index, profitability, and energy consumption. Subsequently, an optimal task schedule is determined using a discrete Keplerian optimization algorithm in combination with Q-learning. Finally, the effectiveness of the proposed methodology and model is demonstrated through numerical experiments. The outcomes are summarized as follows:

- (1) Considering the dual nature of human-machine collaboration and partial disassembly in modern industrial assembly lines, a methodology for human-machine collaboration in partial disassembly lines is proposed. From the perspective of designing a comprehensive disassembly line, three objectives are optimized in terms of productivity, economy and environment: workload balance, profitability and energy efficiency. In addition, hazardous and complex disassembly tasks in the disassembly line are allocated separately to robotic workstations and human

workstations to ensure personnel safety and improve disassembly efficiency. At the same time, common unattributed parts are assigned to robotic workstations by model constraints to reduce personnel workload and labor costs. The workload of the company's employees is prioritized, concurrently with ensuring personnel safety and efficiency, and enhancing the company's competitiveness.

- (2) Numerical findings demonstrate that QLS-KOA exhibits superior performance in addressing HRC PDLBP issues across varying magnitudes and anticipated cycle time constraints. Furthermore, the algorithm's superiority becomes increasingly evident as the dataset expands.
- (3) The HRC-PDLBP and QLS-KOA proposed were effectively implemented in the Tesla Model S power battery module, which has the most complex structure in the electric vehicle market. Research outcomes substantiate that QLS-KOA proficiently produces optimized solutions for disassembly operations, yielding elevated profits, minimized energy consumption, and expeditious task fulfillment. This revelation carries substantial pragmatic implications for the realm of retired power batteries.
- (4) For small, medium, and large sample scenarios, the outcomes of the QLS-KOA algorithm exhibit statistical significance compared to the other four comparative algorithms.

This paper addresses disassembly line balancing within deterministic environments. However, real-world scenarios often involve worn, corroded, or damaged disassembly components, leading to increased uncertainty during disassembly operations. Hence, future studies will incorporate these factors to investigate the challenge of disassembly lines for uncertain environments, thus augmenting the practical relevance and widespread applicability of the proposed model.

CRedit authorship contribution statement

Zenan Tan: Writing – original draft, Methodology. **Baigang Du:** Software, Investigation. **Kaipu Wang:** Validation, Funding acquisition. **Shunsheng Guo:** Supervision, Resources. **Jun Guo:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation of China (52305552; 51705386); China Scholarship Council (201606955091).

Data Availability

Data will be made available on request.

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